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RAIL ASSEMBLY FOR TABLE-MOUNTED MANIPULATOR SYSTEM, AND RELATED DEVICES, SYSTEMS AND METHODS

Abstract

A teleoperable manipulator system comprises a table assembly, a rail coupled to the table assembly, and one or more manipulators coupled to the rail. The table assembly comprises a support column and a platform assembly coupled to the support column and configured to support a body. The platform assembly comprises lateral and longitudinal dimensions defining outer lateral and longitudinal boundaries of the platform assembly. The rail is translatable relative to the support column along a longitudinal dimension of the rail. The manipulators are translatable relative to the rail along the longitudinal dimension of the rail. The rail is configured to remain within at least the outer lateral boundaries of the platform assembly throughout all motion of the rail relative to the support column.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/336,773, filed Apr. 29, 2022, which is incorporated by reference herein in its entirety.

FIELD

[0002] Aspects of this disclosure relate generally to table mounted manipulator systems. In particular, aspects of the disclosure relate to rail assemblies for movably mounting manipulators to a table, such as a medical system table for supporting a patient. Related devices, systems, and methods also are disclosed.

INTRODUCTION

[0003] Computer-assisted manipulator systems (“manipulator systems”), sometimes referred to as robotically assisted systems or robotic systems, can include one or more manipulators that can be operated with the assistance of an electronic controller (e.g., computer) to move and control functions of one or more instruments when coupled to the manipulators. A manipulator generally includes mechanical links connected by joints. An instrument is removably couplable to (or permanently coupled to) one of the links, typically a distal link of the plural links.

[0004] In some computer-assisted manipulator systems, the manipulators are attached to a manipulator support structure (e.g., a patient side cart) that is separate from a support structure that supports a patient or workpiece. In other manipulator systems, the manipulators are attached directly to the support structure that supports the patient or workpiece, e.g., to an operating table. This support structure that supports the patient or workpiece may be referred to herein as a “table assembly” or “table” to simplify the description. Manipulator systems in which the manipulators are mounted to the table assembly may be referred to herein as table-mounted manipulator systems.

[0005] Table-mounted manipulator systems pose certain challenges. The space around a table assembly may need to be occupied with various pieces of equipment and/or personnel during the performance of various tasks that make up a medical procedure. Moreover, the space constraints around the table can vary depending on the medical procedure being performed, with some tasks (such as transferring a patient to the table, draping manipulators, etc.) benefiting from or being facilitated by a large amount of open space around the table. In manipulator systems including movable patient-side carts, such open space around the table can be obtained by moving the patient-side cart away from the table intended to support the patient. However, in table-mounted manipulator systems, moving the manipulators out of the way when space is desired around the table poses challenges as the manipulators are either affixed to the table or at least not practical to remove from the table during a particular stage of a medical procedure. Thus, it can be challenging to avoid interference between the manipulators and the other entities in the space around the table in a table-mounted manipulator system.

[0006] By way of example, while transferring a patient from a gurney to the table, it may be desired for a long side of the table to be substantially free of obstructions so as to allow the gurney to be placed flush with the table to ease the transfer of patient from gurney to table. But when

manipulators are mounted to this long side of the table, the needed open space can be difficult to achieve as the manipulators might get in the way. As another example, while preparing a patient on the table, a number of personnel and equipment may need to be close to the patient and may occupy much of the space around the table and therefore the manipulators mounted to the table can get in the way of personnel and hinder the preparations. As another example, covering the manipulators with a sterile drape can be easier when there is a substantial amount of free space around the manipulator, but when the manipulators are attached to the table assembly it can be difficult to obtain such space due to the proximity of the table.

[0007] In addition, some table assemblies can switch between multiple spatial configurations, such as by lowering or raising sections of a multi-section table, and it can be challenging to accommodate such changes in configuration of the table assembly when the manipulators are mounted to the table assembly. Another challenge with table-mounted manipulators is that the positioning of the manipulators relative to a patient can be limited due to the manipulators being attached to the table, thus making it more difficult to reach certain portions of a patient or certain desired angles during a medical procedure.

[0008] One way to avoid the manipulators becoming an obstruction or otherwise interfering with a task is to place the manipulators in a stowed state during a stage that requires more space around the table. In the stowed state, the manipulators are generally compacted (e.g., folded) and placed in a stowed location, such as under a platform of the table assembly. However, stowing of the manipulators to make sufficient space for a task may not always be feasible, as some tasks require or benefit from the manipulators being in a deployed configuration. Moreover, even when the task being performed does not require the manipulators to be in a deployed configuration, in some circumstances stowing the manipulators during the task can be undesirable.

[0009] For example, stowing the manipulators can compromise the sterility of the manipulators, which can have various undesired effects. The stowing of the manipulators can compromise the sterility of the manipulators because the stowed location (e.g., under the table) is generally not within the sterile field established around the operating table. The sterile field is a region in which any exposed surfaces of objects in the region are maintained in a sterile condition (e.g., a condition substantially free from contaminants, such as biological pathogens, dusts, oils, etc.) and non-sterile surfaces are covered by a sterile barrier. Thus, because stowing the manipulators can compromise their sterility, the manipulators would need to be sterile (e.g., covered with a sterile drape) after completion of the task for which space around the table was needed and once the manipulators have been returned from the stowed state into the sterile field.

[0010] But such sterilization (e.g., draping) of the manipulators after completion of the task can be undesirable in some circumstances. For example, when the task involves the transferring of a patient to the table and/or preparation of the patient on the table, draping the manipulators after completion of the task will cause a delay between when the patient is ready and when subsequent tasks of the overall medical procedure can be performed. Such a delay is generally undesirable, as once the patient has been prepared for the procedure (e.g., transferred to the table) it is generally desirable to proceed with and finish the overall medical procedure as quickly as is feasible. On the other hand, prior to the patient being prepared for the procedure, time is less constrained and delays are generally more acceptable.

[0011] Thus, it is generally desired to perform as many tasks as possible prior to the preparation of the patient so as to minimize the time the patient is on the table. Another reason it can be desirable to drape the manipulators prior to transferring the patient to the table is that it is generally easier to drape the manipulators when the patient is not present, as the personnel may have more room to maneuver at that time. Another reason to drape the manipulators prior to transferring the patient to the table is to avoid having non-sterile objects in the vicinity of a sterile patient, even if only temporarily (e.g., while they are in the process of being draped). Thus, even when stowing of the manipulators to free up space is possible, the stowing of the manipulators is not always be an

acceptable option to free up space for a given stage of an overall procedure.

[0012] Another way to mitigate some of the challenges noted above is to configure the system to allow relative movement between the manipulators and the table assembly. However, mechanisms for providing such relative motion between the manipulator and table generally provide a range of motion that is insufficient to allow the manipulators to move fully out of the way, and those mechanisms that do provide a sufficient range of motion are generally complicated, costly, and/or impractical and tend to themselves obstruct and interfere with other equipment (e.g., due to the mechanism protruding into spaces that would otherwise be occupied by the other equipment). For example, operating tables generally have accessory rails disposed along the side surfaces of the table, with the accessory rails being configured to hold various auxiliary equipment mounted thereto, and mechanisms for providing relative motion of manipulators relative to the table often extend into spaces that would be occupied by such auxiliary equipment attached to accessory rails of the table, thus interfering with usage of such auxiliary equipment. Generally speaking, the wider the range of motion for the manipulators provided by a mechanism, the more likely it is that the mechanism will interfere with the other equipment.

[0013] Thus, it can be difficult to provide a simple, cost effective, and practical mechanism that both enables the manipulators to move relative to the operating table with a sufficient range of motion and does not obstruct or otherwise interfering with the other equipment around the table. Accordingly, a need exists for improved table mounted manipulator systems, in particular systems with improved mechanisms for mounting the manipulators to the table.

SUMMARY

[0014] Various embodiments of the present disclosure can solve one or more of the above-mentioned problems and/or can demonstrate one or more of the above-mentioned desirable features. Other features and/or advantages may become apparent from the description that follows.

[0015] Some embodiments relate to a teleoperated manipulator system. The system can include a table assembly that includes a support column and a platform assembly coupled to the support column and configured to support a body. The platform assembly has lateral and longitudinal dimensions defining outer lateral and longitudinal boundaries of the platform assembly. The system can also include a rail coupled to the table assembly. The rail can be translatable relative to the support column along a longitudinal dimension of the rail. The system can further include one or more manipulators can be coupled to the rail and translatable relative to the rail along the longitudinal dimension of the rail. The rail can be configured to remain within at least the outer lateral boundaries of the platform assembly throughout all motion of the rail relative to the support column.

[0016] In some embodiments, the rail is configured to remain within the outer longitudinal boundaries of the platform assembly throughout all motion of the rail relative to the support column. In some embodiments, the platform assembly includes a plurality of sections movable relative to one another to change a configuration of the platform assembly between a plurality of configurations. In some embodiments, the rail is movable between a plurality of positions to allow the platform assembly to move between the plurality of configurations without interference from the rail. In some embodiment, the plurality of sections include a first end section, one or more middle sections, and a second end section consecutively positioned along the longitudinal dimension of the platform assembly, and each of the first and second end sections are independently pivotable relative to the one or more middle sections.

[0017] In some embodiments, the system further includes one or more first carriages, each first carriage movably coupling one of the manipulators to the rail, each first carriage being drivable to cause translation of the manipulator coupled thereto relative to the rail along the longitudinal dimension of the rail, and also includes one or more second carriages movably coupling the rail to the table assembly, each second carriage being drivable to cause translation of the rail relative to the platform assembly and the support column. In some embodiments, the system can include a

control system communicably connected to the first and second carriages, and the control system can be configured to drive the first and second carriages to position the rail and the manipulators. [0018] In some embodiments, the plurality of sections can include a first section and a second section positioned adjacent to and pivotable relative to the first section, wherein in a first state the second section can be declined relative to the first section and at least a portion of the second section can occupy a given region of space, wherein the rail is translatable relative to the support column between a first position and a second position, in the first position a portion of the rail occupies the given region of space, in the second position the rail does not occupy the given region of space.

[0019] In some embodiments, the system includes a control system configured to, in response to determining a transition to the first state is going to occur, cause the rail to move from the first position to the second position. In some embodiments, the system can include one or more carriages, each carriage movably coupling one of the manipulators to the rail, each carriage being drivable in translation along the longitudinal dimension of the rail so as to cause translation of the manipulator coupled thereto relative to the rail along the longitudinal dimension of the rail.

[0020] In some embodiments, the system can include a drive mechanism operably coupled to drive translation of the one or more carriages along the longitudinal dimension of the rail. The drive mechanism can include a lead screw; one or more follower nuts engaged with the lead screw and coupled with respectively corresponding ones of the one or more carriages; and one or more actuators operably coupled to the follower nuts and configured to rotate the one or more follower nuts relative to the lead screw. One or more rail mounting carriages can movably couple the rail to the platform assembly, the one or more rail mounting carriages being drivable to cause translation of the rail relative to the platform assembly and the support column. In some embodiments, the system can further include a second lead screw; one or more second follower nuts engaged with the second lead screw and coupled with respectively corresponding ones of the one or more rail mounting carriages, and one or more second actuators operably coupled to the second follower nuts to drive translation of respectively corresponding ones of the one or more rail mounting carriages relative to the rail by driving rotation of the respectively corresponding second follower nuts relative to the second lead screw. The one or more rail mounting carriages can movably couple to the fixed rail and be translatable relative to the platform assembly along the fixed rail.

[0021] In some embodiments, the drive mechanism can include one or more cables coupled with respectively corresponding ones of the one or more carriages; one or more capstans, where the one or more cables are at least partially wrapped around the one or more capstans, respectively; and one or more actuators respectively operably coupled to the one or more capstans and configured to rotate the one or more capstans pay in or pay out, based on a direction of rotation, the one or more cables relative to the one or more capstans.

[0022] In some embodiments, the one or more carriages can include a first carriage and a second carriage and the system can further comprise a rail mounting carriage movably coupling the rail to the platform assembly. In some embodiments, the one or more cables include a first cable coupled with the first carriage, a second cable coupled with the second carriage, and a third cable coupled with the rail mounting carriage, and the one or more capstans comprises a first capstan, a second capstan, and a third capstan. The first cable can be at least partially wrapped around the first capstan, the second cable can be at least partially wrapped around the second capstan.

[0023] In some embodiments, the platform assembly is tiltable relative to the support column and the rail is configured to tilt along with the platform assembly relative to the support column. In some embodiments, the platform assembly can include a plurality of sections movable relative to one another to change a configuration of the platform assembly between a plurality of configurations, where a first section of the plurality of sections is coupled to the support column and is tiltable relative to the support column, and the rail can be coupled to the first section. The outer lateral boundaries of the platform assembly can be defined by outermost edges of the

platform assembly along the longitudinal side excluding the accessory rail. In some embodiments, the rail is movably coupled to the rail mounting carriage and is configured to move along the rail mounting carriage to cause translation of the rail relative to the support column along the longitudinal dimension of the rail.

[0024] In some embodiments, the system includes a drive mechanism operably coupled to drive movement of the rail along the rail mounting carriage. In some embodiments, the drive mechanism can include a lead screw; a follower nut engaged with the lead screw and coupled with the rail and the rail mounting carriage, and an actuator operably coupled to the follower nut and configured to rotate the follower nut relative to the lead screw. In some embodiment, the drive mechanism can include a cable coupled with rail mounting carriage; a capstan, the cable at least partially wrapped around the capstan; and an actuator operably coupled to drive rotation of the capstan, where rotation of the capstan pays in or pays out, based on a direction of rotation, the cable relative to the capstan.

[0025] Some embodiments relate to a rail assembly for mounting manipulators to a table assembly. In some embodiments, the rail assembly can include a rail having a longitudinal dimension; a rail mounting carriage movably engaged with the rail and configured to couple the rail to a table assembly; and first and second manipulator mounting carriages movably engaged with the rail at an opposite side of the rail as the rail mounting carriage. In some embodiments, the rail can further include a drive mechanism housed in an interior of the rail. The drive mechanism can include a first capstan assembly, a first cable operably coupled to the first capstan assembly and to the first manipulator carriage, the first cable passing through the second manipulator carriage; a second capstan assembly, a second cable operably coupled to the second capstan assembly and the second manipulator carriage, the second cable passing through the first manipulator carriage; and a third capstan assembly and a third cable operably coupled to the third capstan assembly and the rail mounting carriage. In some embodiments, the first, second, and third capstan assemblies can be arranged in a series along the longitudinal dimension of the rail.

Description

BRIEF DESCRIPTION OF TUE DRAWINGS

[0026] The present disclosure can be understood from the following detailed description, either alone or together with the accompanying drawings. The drawings are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate one or more embodiments of the present teachings and together with the description explain certain principles and operation. In the drawings:

[0027] FIG. 1A is a schematic side view of an embodiment of a table-mounted manipulator system in a first state.

[0028] FIG. 1B is a schematic side view of the table-mounted manipulator system of FIG. 1A in a second state.

[0029] FIG. 1C is a schematic side view of the table-mounted manipulator system of FIG. 1A in a third state.

[0030] FIG. 2 is a schematic front view of the table-mounted manipulator system of FIG. 1A.

[0031] FIG. 3 is a perspective view of another embodiment of a table-mounted manipulator system.

[0032] FIG. 4A is a perspective view of the table-mounted manipulator system of FIG. 3.

[0033] FIG. 4B is an enlarged perspective view of a portion of the table-mounted manipulator system of FIG. 4A with a rail housing made transparent.

[0034] FIG. 5A is a side view of the table-mounted manipulator system of FIG. 4A in a first state.

[0035] FIG. 5B is a side view of the table-mounted manipulator system of FIG. 4A in a second state and illustrating a third state in dashed lines.

[0036] FIG. 5C is a side view of the table-mounted manipulator system of FIG. 4A in a fourth state and illustrating a fifth state in dashed lines.

[0037] FIG. 5D is a side view of the table-mounted manipulator system of FIG. 4A in a sixth state and illustrating a seventh state in dashed lines.

[0038] FIG. 6 is a perspective view of another embodiment of a table-mounted manipulator system in a first state.

[0039] FIG. 7 is a side view of the table-mounted manipulator system of FIG. 6 in a second state.

[0040] FIG. 8A is a perspective view of a table assembly with a rail assembly in accordance with another embodiment.

[0041] FIG. 8B is a side view of a portion of the rail assembly of FIG. 8A shown in isolation.

[0042] FIGS. 9A and 9B are a perspective side views of FIG. 8B with the rail housing removed to show translation drive mechanisms of the rail assembly, with the view of FIGS. 9A and 9B taken from opposite sides.

[0043] FIG. 10 is an isolated view of an embodiment of an actuator and capstan portion of a translation drive mechanism.

DETAILED DESCRIPTION

[0044] As noted above, there can be certain challenges arising from having manipulators attached to a table assembly in a table-mounted manipulator system, such as challenges associated with the manipulators, personnel, and/or a table obstructing or interfering with one another during various stages of a medical procedure.

[0045] To address challenges with table mounted manipulator systems, various embodiments disclosed contemplate a table-mounted manipulator system including a table assembly and a rail that is coupled to the table assembly. The rail can be translatable relative to and along a longitudinal dimension of the table assembly. The rail supports one or more manipulators, which are translatable along the rail. The translation of the rail relative to the table assembly combined with the translation of the manipulators along the rail allows for a relatively wide range of motion of the manipulators relative to the table assembly, especially when compared with the relatively narrower range of motion that would be provided if the manipulators were to translate along a similarly dimensioned but stationary rail. The wide range of motion can beneficially allow the manipulators to be moved more fully out of the way when desired, as well as allowing more flexibility in positioning instruments supported by the manipulators.

[0046] Moreover, in various embodiments disclosed herein the rail is configured to remain within outer lateral and longitudinal boundaries of a platform of the table assembly throughout all motion of the rail (i.e., an entirety of the rail is within the boundaries of the platform of the table assembly). This helps to ensure that the rail does not itself become an obstruction or interfere with other equipment, such as auxiliary equipment mounted to an accessory rail. Thus, in embodiments disclosed herein, a large range of motion is provided for the manipulators while also avoiding having the mechanism that provides the motion interfere with other equipment.

[0047] FIGS. 1A-D illustrate an embodiment of a table-mounted manipulator system **100** (“system **100**”). The system **100** includes a table assembly **101**, at least one rail assembly **120** coupled to the table assembly, and one or more manipulators **140** coupled to each rail assembly **120**. Each manipulator **140** can carry one or more instruments **150**, which can be removably or permanently mounted thereon. As shown in FIG. 1A, the system **100** also can include a control system **1006**, a user input and feedback system **1004**, and/or an auxiliary system **1008**. In some embodiments, the system **100** is configured as a computer-assisted, teleoperable medical system, in which case table assembly **101** can be configured to support a patient (not shown) and the instruments **150** are medical instruments. The system **100** in this configuration is usable, for example, to perform any of a variety of medical procedures, such as surgical procedures, diagnostic procedures, imaging procedures, therapeutic procedures, etc.

[0048] Moreover, the system **100** when configured as a teleoperable medical system need not

necessarily be used on a living human patient. For example, a non-human animal, a cadaver, tissue-like materials used for training purposes, and so on, can be supported on the table assembly **101** and worked on by system **100**. In other embodiments, the system **100** is configured as a computer-assisted teleoperable system for use in non-medical contexts, in which case the table assembly **101** can be configured to support an inanimate workpiece (something being manufactured, repaired, tested, etc.) and the instruments **150** can be non-medical instruments, such as industrial instruments.

[0049] As shown in FIG. **1A**, the table assembly **101** includes a platform assembly **110** configured to support the patient or inanimate workpiece, a support column **102** coupled to and supporting the platform assembly **110**, and a base **105** coupled to the support column **102**. The base can be configured to contact the ground or other surface upon which the table assembly **101** rests to provide stability for the table assembly **101**. In some embodiments, the base **105** is omitted. In some embodiments, the base **105** includes mobility features, such as wheels, tracks, or other such features (not shown), to allow movement of the table assembly **101** along the ground or other surface. In FIGS. **1A-2**, the support column **102** is illustrated as a single vertical columnar part to simplify the discussion, but the support column **102** could take any desired shape and could include any number of parts. For example, the support column **102** can include horizontal support structures (not illustrated) such as beams, rails, etc. to couple the platform assembly **110** to a vertical portion of the support column **102**. Moreover, in various embodiments, the support column **102** can be telescoping and configured to extend and contract in height.

[0050] The platform assembly **110** includes one or more platform sections **103** to support the patient or workpiece. The platform sections **103** each have a support surface configured to contact and support the patient or workpiece. In some embodiments multiple platform sections **103** are used and the platform sections **103** are arranged in series to support different portions of the patient or workpiece.

[0051] For example, in the embodiment illustrated in FIG. **1A**, the platform assembly **110** includes a first end section **103_1**, one or more middle sections **103_2**, and a second end section **103_3**, with the one or more middle sections **103_2** being arranged between the two end sections **103_1** and **103_3**. In some embodiments, the first end section **103_1** is configured to support a head of the patient, the second end section **103_3** is configured to support the feet and/or legs of the patient, and the one or more middle sections **103_2** are configured to support a torso and/or other portions of the patient. For convenience, the side of the platform assembly **110** that is near the first end section **103_1** (e.g., a left side in the orientation shown in FIG. **1A**) will be referred to herein as a “head” of the platform assembly **110** (or “head side” or “head end”) and the side of the platform assembly **110** that is near the second end section **103_3** (e.g., a right side in the orientation shown in FIG. **1A**) will be referred to herein as a “foot” of the platform assembly **110** (or “foot side” or “foot end”), but this is merely an arbitrary convention chosen herein for convenience of description and is not intended to limit the configuration or usage of the table assembly **101** (e.g., a patient's head could be positioned at the “foot” side of the platform assembly **110** if desired, and vice versa). The relative positions of two components or of two portions of a single component also can be described using “head” and “foot” (e.g., a “head end” and a “foot end” of a rail **121**) with “head” referring to the component or portion that is relatively closer to the head end of the platform assembly **110** and “foot” referring to the component or portion that is relative closer to the foot end of the platform assembly **101**.

[0052] In other embodiments, different numbers and arrangements of platform sections **103** are used, including one, two, four, or more platform sections **103**. In some embodiments, one or more of the platform sections **103** can be movable relative to other platform sections **103** and/or relative to the support column **102**. For example, in some embodiments, some or all of the platform sections **103** are coupled to adjacent platform sections **103** and/or to the support column **102** by rotatable joints such that at least some of the platform sections **103** can tilt relative to one another

and/or relative to the support column **102**. The platform assembly **110** can also be movable as a whole relative to the support column **102**, as described in greater detail below.

[0053] The platform assembly **110** has a longitudinal dimension **198** (see FIG. 1A), a lateral dimension **199** (see FIG. 2) orthogonal to the longitudinal dimension, and a thickness or height dimension (not labeled) orthogonal to both the longitudinal and lateral dimensions. As used herein, the longitudinal dimension **198** refers to a dimension of greatest extent of the platform assembly **110**, e.g., when all of the platform sections **103** of the platform assembly are fully extended and all are oriented with their support surfaces roughly aligned in a same plane with one another (or when as close to this state as possible) so as to collectively form a combined support surface that is substantially planar with potentially small gaps between adjacent platform sections **103**.

[0054] The longitudinal direction extends along the head-to-foot and vice-versa directions. In general, the longitudinal and lateral dimensions **198** and **199** of the platform assembly **110** and the support surfaces of the platform sections **103** are oriented roughly parallel to the ground or other surface on which the table assembly **101** is supported when the platform assembly **110** is in a neutral configuration. For example, in FIGS. 1A-2 the longitudinal dimension **198** is parallel to the x-direction and the lateral dimension **199** is parallel to the y-direction, with the x- and y-directions being parallel to the ground or other surface the table assembly **101** rests upon. Thus, in FIGS. 1A-2, the thickness dimension is parallel to the z-direction, which is perpendicular to the ground or other surface.

[0055] However, one of ordinary skill in the art would understand that the platform assembly **110** as a whole and/or individual platform sections **103** thereof do not necessarily have to be parallel to the ground, and that one or both of the longitudinal and/or lateral dimensions **198** and **199** can be tilted relative to the ground in various configurations through which the platform assembly **110** and/or platform section **103** can be movable, including in a neutral configuration in some cases. As used herein, a longitudinally extending side of the platform assembly refers to either of the sides that are parallel to the longitudinal dimension **198**. A laterally extending side of the platform assembly refers to either of the sides that are parallel to the lateral dimension **199**.

[0056] At least one of the platform sections **103** is directly coupled to and supported by the support column **102**. The remaining platform sections **103** can be coupled directly to the support column **102** or indirectly to the support column **102** via a chain of one or more intervening platform sections **103**. For example, in some embodiments a main platform section **103** (e.g., a middle section **103_2**) is coupled to and directly supported by the support column **102** and the others of the platform sections **103** (e.g., end sections **103_1** and **103_3**) are coupled to the main platform section **103** or to another platform section **103**. As another example, in some embodiments multiple platform sections **103** (all in some embodiments) are coupled directly to the support column **102** and not to another platform section **103**.

[0057] In some embodiments, some (all, in some cases) of the above-described parts of the table assembly **101** are movable relative to one another. For example, in some embodiments the platform assembly **110** as a whole can be moved relative to the support column **102**, such as by tilting around a horizontal axis, swiveling around a vertical axis, translating vertically along the support column **102**, translating horizontally relative to the support column **102**, and so on. In some embodiments, such movement of the platform assembly **110** as a whole is provided by one or more joints that couple a main platform section **103** (e.g., a middle section **103_2**) to the support column **102**. Furthermore, as already noted above, individual platform sections **103** can be movable relative to one another and relative to the support column **102** as well, which can be facilitated by joints coupling the platform sections **103** to the support column **102** or to adjacent platform sections **103**.

[0058] In some embodiments, the platform assembly **110** also includes one or more accessory rails **104**. The accessory rails **104** can be configured to receive accessory devices removably mounted thereon, such as arm supports, leg supports, body restraints, bed extensions, various clamps for surgical retractors, and other device holders. In some embodiments, the accessory rails **104** adhere

to industry standard specifications familiar to those of ordinary skill in the art to allow compatibility with accessory devices compliant with the standard. The accessory rails **104** can be attached along longitudinally extending sides of one or more of the platform sections **103**. One or more openings can be defined between an accessory rail **104** and the side of the platform section **103** to which the accessory rail **104** is attached and portions of accessories mounted to the accessory rail **104** can be inserted through the openings.

[0059] As noted above, the system **100** includes one or more manipulators **140**. FIGS. **1A-2** illustrate two manipulators **140**, but any number of manipulators **140** can be included (such as, for example, one, two, three, or more manipulators mounted to each rail assembly **120**, as described in further detail below). A manipulator includes a kinematic structure of links coupled together by one or more joints. The manipulator **140** is movable through various degrees of freedom of motion provided by the joints, thus allowing an instrument **150** mounted thereon to be moved relative to the worksite. For example, some joints can provide for rotation of links relative to one another, other joints can provide for translation of links relative to one another, and some can provide for both rotation and translation. Some or all of the joints can be powered joints, meaning a powered drive element can control movement of the joint through the supply of motive power. Such powered drive elements include, but are not limited to, for example, electric motors, pneumatic or hydraulic actuators, etc. Additionally, some joints can be unpowered joints. The specific number and arrangement of links and joints is not limited. The more links and joints are included, the greater the degrees of freedom of movement of the manipulator **140**. A proximal end portion of each manipulator **140** is movably coupled to the table assembly **101** via a rail assembly **120**, as described in further detail below.

[0060] The instrument **150** can be removably mounted or permanently affixed to the manipulator **140** via an interface. The instruments **150** can include any tool or instrument, including for example industrial instruments and medical instruments (e.g., surgical instruments, imaging instruments, diagnostic instruments, therapeutic instruments, etc.). In embodiments in which the instrument **150** is removably mountable to the manipulator **140**, the manipulator **140** can include an instrument manipulator mount (not illustrated) to which the instrument can be removably coupled. The instrument manipulator mount can be located, for example, at a generally distal end portion of the manipulator **140**. The manipulator mount has an interface (not illustrated) including output couplers (not illustrated) to engage (directly or indirectly via an intermediary) with input couplers (not illustrated) of the instrument **150** to provide driving forces or other inputs to the mounted instrument **150** to control various degree of freedom movement and/or other functionality of the instrument **150**, such as moving an end-effector of the instrument, opening/closing jaws, driving translation and/or rotation of a variety of components of the instrument, delivery of substances and/or energy from the instrument, and various other functions those of ordinary skill in the art are familiar with.

[0061] The output couplers can be driven by actuators (e.g., electrical servo-motors, hydraulic actuators, pneumatic actuators) with which those of ordinary skill in the art have familiarity. An instrument sterile adaptor (ISA) can be disposed between the instrument **150** and the instrument manipulator mount interface to maintain sterile separation between the instrument **150** and the manipulator **140**. The instrument manipulator mount can also include other interfaces (not illustrated), such as electrical interfaces to provide and/or receive electrical signals to/from the instrument **150**. In some embodiments, the system **100** can include flux delivery transmission capability as well, such as, for example, to supply electricity, fluid, vacuum pressure, light, electromagnetic radiation, etc. to the end effector. In other embodiments, such flux delivery transmission can be provided to an instrument through another auxiliary system **1008**, described further below and as those of ordinary skill in the art would be familiar with in the context of computer-assisted, teleoperated medical systems.

[0062] In some embodiments, the manipulators **140** can be similar to the manipulators described in

U.S. Patent Application No. 63/336,778 entitled “NESTING PROXIMAL LINKS FOR TABLE MOUNTED MANIPULATOR SYSTEM, AND RELATED DEVICES, SYSTEMS AND METHODS,” first named inventor Brain Lambrecht, and in U.S. Patent Application No. 63/336,840 entitled “TABLE-MOUNTED MANIPULATOR SYSTEM, AND RELATED DEVICES, SYSTEMS AND METHODS,” first named inventor Steven Manuel, both filed on a date even herewith, or those described in, for example, U.S. Pat. No. 9,358,074 (filed May 31, 2013) to Schena et al., entitled “Multi-Port Surgical Robotic System Architecture,” U.S. Pat. No. 9,295,524 (filed May 31, 2013) to Schena et al., entitled “Redundant Axis and Degree of Freedom for Hardware-Constrained Remote Center Robotic Manipulator,” and U.S. Pat. No. 8,852,208 (filed Aug. 12, 2010) to Gomez et al., entitled “Surgical System Instrument Mounting,” the contents of each of which are incorporated herein by reference in their entirety. Various other embodiments of manipulators can include those as configured as part of the medical systems that are part of various da Vinci® Surgical Systems, such as the da Vinci X®, da Vinci Xi®, and da Vinci SP systems, commercialized by Intuitive Surgical, Inc., of Sunnyvale, California.

[0063] As shown in FIGS. 1A-2, the manipulators **140** are coupled to the table assembly **101** via the at least one rail assembly **120**. In some embodiments, multiple similar rail assemblies **120** are provided, for example one for each longitudinally extending side of the platform assembly **110**. For example, in some embodiments, a first rail assembly **120** can be provided at a first longitudinally extending side of the platform assembly **110** and a second rail assembly **120** can be provided at a second longitudinally extending side of the platform assembly **110**. In such embodiments with multiple rail assemblies **120**, manipulators **140** can be coupled to the rail assemblies **120** in any number or combination. With the rail assemblies **120** positioned along multiple sides of the platform assembly **120**, the manipulators **140** too are positioned along multiple sides of the platform assembly **140**. The description below will describe one rail assembly **120** to simplify the description, but the other rail assemblies **120** (if present) can be configured similarly.

[0064] The rail assembly **120** includes a movable rail **121** (“rail **121**”), one or more movable manipulator carriages **126** (two being shown in the embodiment of FIGS. 1A-2) coupled to the rail **121** and to the manipulators **140** (e.g., at least one carriage **126** can be provided per manipulator **140**), and one or more rail mounting carriages **127** coupled to the rail **121** and the table assembly **101**. In some embodiments, the rail assembly **120** further includes a fixed rail **124**, which can be coupled between the rail mounting carriages **127** and the table assembly **101** (either to the platform assembly **110** or alternatively, as depicted by the dashed structure **124'** in FIG. 2, to the support column **102**), while in other embodiments the rail mounting carriages **127** can be coupled directly to the table assembly **101**.

[0065] Each movable manipulator carriage **126** is moveable along a longitudinal dimension **197** of the rail **121** and couples a respectively corresponding one of the manipulators **140** to the rail **121** such that the manipulators **140** can translate relative to the rail **121** along the longitudinal dimension **197** of the rail **121**. In FIGS. 1A-2, one movable manipulator carriage **126** is shown per manipulator **140**, but multiple movable manipulator carriages **126** could be provided to operably couple to and support a given manipulator **140**. In some embodiments, the longitudinal dimension **197** of the rail **121** is parallel to the longitudinal dimension **198** of the platform assembly **110** (e.g., parallel to the x-axis) in a neutral configuration of the platform assembly **110**, as shown in FIG. 1A. The movable rail **121** includes a first set of engagement features **122** configured to engage with complementary engagement features **128** of the movable manipulator carriages **126**. For example, as shown in FIG. 2, the first set of engagement features **122** of the rail **121** includes a track including flanges extending along the longitudinal dimension **197**, and the complementary engagement features **128** of the carriages **126** are configured to engage and ride along the flanges of the first set of engagement features **122**. The first set of engagement features **122** can also include a track including grooves in which the complementary engagement feature **128** are received.

[0066] Any other type of complementary engagement features that when engaged allow for relative translation can be used as the complementary engagement features **122** and **128**, and those having ordinary skill in the art are familiar with various complementary engagement features that are used in rail and carriage systems. In some embodiments, the first set of engagement features **122** and/or the complementary engagement features **128** include bearing devices configured to reduce friction to facilitate easier translation, such as wheels, balls, plain bearing surfaces coated or otherwise provided with a low friction material, and other friction reducing mechanisms.

[0067] The one or more rail mounting carriages **127** couple the rail **121** to the table assembly **101** such that the rail **121** can translate relative to the platform assembly **110** and/or support column **102** along directions of the longitudinal dimension **197** of the rail **121**. This translation between the rail **121** and the platform assembly **110** and/or support column **102** is provided at least in part by a movable coupling between the rail mounting carriages **127** and the rail **121**, such that the rail **121** and the second carriages **127** can translate relative to one another along the directions of the longitudinal dimension **197** of the rail **121**. In addition, in some embodiments, the rail mounting carriages **127** are also movably coupled to the table assembly **101** such that the second carriages **127** are translatable relative to the table assembly **101** along the directions of the longitudinal dimension **197** of the rail **121**. Thus, in these embodiments, translation of the rail **121** relative to the table assembly **101** can occur through either one or both of relative translation between the carriages **127** and the table assembly **101** and relative translation between the carriages **127** and rail **121**.

[0068] For example, in some embodiments a first end of the rail mounting carriage **127** is movably coupled to a fixed rail **124**, which in turn is attached to one of the platform sections **103** or to the support column **102** (the latter shown in dashed at **124'**), and the rail mounting carriage **127** is translatable along the fixed rail **124**. However, in other embodiments the first end of the rail mounting carriage **127** is fixedly attached to or is part of the platform assembly **110** and/or support column **102**, and thus the carriages **127** remain stationary relative to the platform assembly **110** and/or support column **102** (in such embodiments, the fixed rail **124** is omitted).

[0069] Regardless of whether the first end of the rail mounting carriage **127** is fixedly or movably coupled to the platform assembly **110** and/or support column **102**, in some embodiments a second end of the rail mounting carriage **127** is moveably coupled to the movable rail **121** such that the rail **121** and the carriage **127** are translatable relative to one another. In particular, the rail includes a second set of engagement features **123** (e.g., tracks or other engagement features) that engage with complementary engagement features **125** of the carriage **127** to couple the rail **121** to the carriage **127** while allowing translation between the rail **121** and carriage **127**. The movability of the carriages **127** relative to the table assembly **101** can allow for a greater range of motion of the manipulators **140** and/or for a shorter length of the rail **121**, as compared to a configuration in which the rail **121** is fixed relative to the table assembly **101**. This can also enable the rail assembly **120** to more easily be moved out of the way of the platform assembly **110** to avoid interference therewith as the platform assembly **110** and/or individual platform sections **103** thereof are moved through various configurations.

[0070] In other embodiments, the rail mounting carriages **127** are stationary relative to the table assembly **101** and translation of the rail **121** relative to the table assembly **101** occurs solely through relative translation between the carriages **127** and the rail **121**. For example, in some embodiments a first end of a rail mounting carriage **127** is attached to one of the platform sections **103** or to the support column **102** and a second end of the second carriage is moveably attached to the movable rail **121**. One rail mounting carriage **127** is shown in FIGS. **1A-2** for ease of description, but any number could be used.

[0071] As noted above, in some embodiments the one or more rail mounting carriages **127** are coupled (directly or via an intermediary such as the fixed rail **124**, **124'**) to one of the platform sections **103** or to the support column **102**. Which structure the carriages **127** are coupled to can

make a difference in embodiments in which the platform assembly **110** as a whole is movable relative to the support column **102**, for example by tilting relative to the support column **102**. In embodiments in which the rail mounting carriages **127** are coupled to one of the platform sections **103** (e.g., a middle section **103_2**) (either directly or via an intermediary such as fixed rail **124**), when the platform assembly **110** moves relative to the support column **102**, the rail assembly **120** and hence the manipulators **140** coupled thereto move along with the platform assembly **110**. This can allow the manipulators **140** to automatically maintain a set pose and position relative to the platform assembly **110**, and thus relative to a patient supported on the platform assembly, regardless of a configuration of the platform assembly **110** and without having to reposition the manipulators **140**. Moreover, in some circumstances, collision between the platform assembly **110** and the rail assembly **120** due to motion of the platform assembly **110** can be avoided as they both move together.

[0072] In other embodiments in which the rail mounting carriages **127** are coupled to the support column **102** (either directly or via an intermediary such as the fixed rail **124'**, when the platform assembly **110** moves relative to the support column **102**, the rail assembly **120** and hence the manipulators **140** coupled thereto remain with the support column **102** and do not move along with the platform assembly **110**. This can allow greater motion of the rail assembly **120** and manipulators **150** relative to the platform assembly **110**. This also can increase the strength of the structure between the manipulators **150** and the support column **102** by reducing the length of the structure between them and reducing the number of connections between them.

[0073] In some embodiments, motors or other actuation devices (not illustrated) are provided to drive the relative translation between the rail **121** and the carriages **126**, between the rail **121** and the carriages **127**, and/or between the carriages **127** and the table assembly **101**. In some embodiments, motors/actuators are housed within the rail **121**. In some embodiments, motors/actuators are housed within the carriages **126** and/or **127**.

[0074] FIGS. **1A-1C** show the rail assembly **120** in various configurations illustrative of a range of motion provided by the rail assembly **120** in some embodiments. In FIG. **1A**, the rail mounting carriage **127** is positioned at a middle portion of the movable rail **121** and thus the movable rail **121** is positioned near a middle portion of the table assembly **101** under the middle section(s) **103_2**. In this configuration, the manipulators **140** can be positioned in a variety of positions, for example, along the middle section(s) **103_2** of the platform assembly **110**, which can be useful for performing a variety of operations. Moreover, in this configuration the rail assembly **120** and manipulators **140** are positioned away from the two end regions (head and foot end regions) of the platform assembly **110**, which can beneficially free up space at these regions and thus allow personnel or other equipment to be positioned at, near, or under the end regions of the platform assembly **110**.

[0075] In addition, in some embodiments, one or both of the end platform sections **103_1** and **103_3** could be lowered (declined) relative to the middle section **103_2** while the rail assembly **120** is in this configuration because in some instances the rail **121** does not protrude into the space that would be occupied by the lowered end sections **103_1** or **103_3** and hinder those section from lowering. Whether the rail **121** protrudes into the space occupied by the end sections **103_1** or **103_3** depends on the sizes (lengths) of the rail **121** and platform assembly **110** as well as on the positioning of the rail **121**.

[0076] In FIG. **1B**, a foot end portion of the movable rail **121** and thus the movable rail **121** is positioned near a foot end of the table assembly **101** under the second end section **103_3**. That is, to reach the state shown in FIG. **1B** from the state illustrated in FIG. **1A**, the rail **121** is translated longitudinally toward the foot end of the table assembly relative to the rail mounting carriage **127** until a head end portion of the rail **121** reaches the carriage **127**. In embodiments in which the rail mounting carriage **127** is also movable relative to the fixed rail **124**, the carriage **127** can also be translated toward the foot end of the fixed rail **124** (both translation of the rail **121** relative to the

second carriage and of the second carriage **127** relative to the rail **124** can occur simultaneously, or in any order, as desired). In this configuration, the manipulators **140** can be positioned, for example, along the longitudinally extending side **109b** of the platform assembly **110** near the second end section **103_3** of the platform assembly **110** and near some portions of the middle section **103_2**. Moreover, because the rail assembly **120** is moved away of the first end section **103_1** in this configuration, the first end section **103_1** can be placed in a declined position relative to the middle section **103_2** without interference and/or personnel or other equipment could be positioned at/near/under the first end section **103_1**.

[0077] Furthermore, in some embodiments, the positioning of the rail **121** shown in FIG. **1B** allows the manipulators **140** to be moved in a longitudinal (x-axis) direction beyond the foot end of the platform assembly **110**, which can enable the manipulators **140** to further be moved in a laterally inward direction (positive y-axis direction) such that the manipulators **140** are positioned along the head end, laterally extending side **109a** of the platform assembly **110** instead of along the longitudinally extending side **109b** of the platform assembly **110**. Thus, in this position the manipulators **140** are out of the way of and do not obstruct the longitudinally extending side **109b** of the platform assembly **110**. Such positioning can be desirable in various stages of a medical procedure, such as, for example, transferring a patient from a gurney to the platform assembly **110**. Moreover, in this position the manipulators **140** remain in the sterile field and thus can be moved out of the way for a task (e.g., patient transfer) while remaining covered with a sterile drape prior to the task (e.g., patient transfer).

[0078] In FIG. **1C**, the rail mounting carriage **127** is positioned at a foot end portion of the movable rail **121** (right end of movable rail **121** in FIG. **1C**) and therefore the rail **121** is positioned near the head end of the table assembly **101** under the first end section **103_1**. That is, to reach the state shown in FIG. **1C** from the state illustrated in FIG. **1A**, the rail **121** is translated longitudinally toward the head end of the table assembly **101** relative to the carriage **127**. In embodiments in which the carriage **127** is also movable relative to the fixed rail **124**, the carriage **127** can also be translated toward the head end of the fixed rail **124** (both translation of the rail **121** relative to the second carriage and of the carriage **127** relative to the rail **124** can occur simultaneously, or in any order, as desired). In this configuration, the manipulators **140** can be positioned, for example, along the head end, longitudinally extending side **109b** of the table assembly **101** near the first end section **103_1** of the platform assembly **110** and near portions of the middle section **103_2**. Moreover, because the rail assembly **120** is moved away of the second end section **103_3** in this configuration, the second end section **103_3** can be placed in a declined position relative to the middle section **103_2** without interference and/or personnel or other equipment could be positioned at/near/under the second end section **103_3**.

[0079] In some embodiments (not illustrated), the positioning of the rail **121** allows the manipulators **140** to be moved in a longitudinal (x-axis) direction beyond the head end of the platform assembly **110** and then in a lateral (y-axis) direction such that the manipulators **140** are positioned along the laterally extending side **109a** of the head end of the platform assembly **110** instead of along the longitudinally extending side **109b** of the platform assembly **110**, similar to the positioning described above in relation to FIG. **1B** except at the head end of the platform assembly rather than the foot end. In some embodiments, the manipulators **140** can be moved beyond both the head and foot ends of the platform assembly **110** in the above-described manner, in other embodiments the manipulators **140** can be moved beyond the head end only, and in still other embodiments can be moved beyond the foot end only.

[0080] Thus, the rail assembly **120** of the embodiment of FIGS. **1A-2** provides a wide range of motion for the manipulators **140**, which can allow the manipulators **140** to be positioned nearly anywhere along the length of the platform assembly **110**. Moreover, not only does the rail assembly **120** provide a wide range of motion for the manipulators **140**, it does so while also minimizing interference between the rail assembly **120** and the platform assembly **110**, personnel, and other

equipment. In particular, the rail assembly is movable between different positions based on the configuration of the platform assembly **110** and/or based on various tasks during a medical procedure that can benefit from the rail **121** positioned out of the way of the portions of the platform assembly **110** (e.g., out of the way of a platform section **103** being lowered), personnel, or other equipment as needed.

[0081] Furthermore, the rail assembly **120** can be configured to remain within an overall envelope **108** defined by the platform assembly **110**, which corresponds to a footprint or peripheral profile of the platform assembly **110** when viewed from above (i.e., a projection of the platform assembly **110** onto the plane of the ground surface), throughout all motion of the rail **121** relative to the platform assembly **110** and/or support column **102** (i.e., an entirety of the rail is within the envelope **108**). This further reduces the degree of interference between the rail assembly **120** and other equipment, as will be described in greater detail below. The envelope **108** includes longitudinal boundaries **108a** and lateral boundaries **108b** defined by outer laterally extending sides **109a** and outer longitudinally extending sides **109b** of the platform assembly **110**. The outer longitudinal and lateral boundaries **108a**, **108b** of the envelope **108** correspond to vertical projections from the aforementioned outer laterally and longitudinally extending sides **109a** and **109b** of the platform assembly **110**, respectively, down to the ground or other surface upon which the table assembly **101** rests. For example, in embodiments in which the platform assembly **110** has a rectangular profile, as shown in FIG. 1A, the longitudinal boundaries **108a** of the envelope **108** correspond to vertical planes that are perpendicular to the longitudinal dimension **198** (i.e., perpendicular to the x-axis) and tangent to the laterally extending sides **109a**, and similarly, as shown in FIG. 2, the lateral boundaries **108b** of the envelope **108** correspond to vertical planes that are perpendicular to the lateral dimension **199** (i.e., perpendicular to the y-axis) and tangent to the longitudinally extending sides **109b**. The envelope **108** can also include upper and lower boundaries (not labeled) corresponding, respectively, to a top surface of the platform assembly **110** and to the ground (or other surface upon which the table assembly **101** rests).

[0082] As shown in FIG. 2, in some embodiments the laterally extending sides **109a** of the platform assembly **110** that defines the longitudinal boundaries **108a** of the envelope **108** is the outermost edge of the platform assembly **110** excluding the accessory rail **104**, if present. Thus, in such embodiments, the envelope **108** defined by the platform assembly **110** is contained laterally inside of the accessory rails **104**, if present. This can be beneficial in some circumstances because it ensures that the rail assembly **120** will not interfere with any accessories that are mounted to the accessory rail **104**. In particular, because the rail assembly **120** remains within the envelope **108** throughout the entire range of motion of the rail **121**, and because the envelope **108** is laterally inside of the accessory rails **104** in these embodiments, the rail **121** or other portions of the rail assembly **120** cannot collide with or otherwise interfere with the accessories. Moreover, even in embodiments in which the rail **104** is not present or when the envelope **108** extends to the edge of the rail **104**, the rail assembly **120** remaining within the envelope **108** of the platform assembly **110** still helps to reduce interference by the rail assembly **120** with other equipment or personnel positioned around the table assembly **101**, as it can be known in advance that the rail assembly **120** will not collide with or otherwise interfere with equipment or personnel that are positioned outside of the envelope **108**.

[0083] Although embodiments are described above in which the envelope **108** includes both longitudinal and lateral boundaries **108a**, **108b** and the rail assembly **120** remains within all of these boundaries throughout its range of motion, in other embodiments the rail assembly **120** can remain within only a subset of these boundaries. For example, in some embodiments, the rail assembly **120** remains within the lateral boundaries **108b**, but not necessarily within the longitudinal boundaries **108a** (e.g., the rail assembly **120** can be moved to be positioned to at least partially extend beyond the longitudinal boundaries **108a** (e.g., beyond a head or foot end of the platform assembly) while remaining within the lateral boundaries **108b** in some embodiments).

[0084] Returning to FIG. 1A, the user input and feedback system **1004**, control system **1006**, and auxiliary system **1008** will be described. Some or all of these components can be provided at a location remote from the table assembly **101**. The user input and feedback system **1004** is operably coupled to the control system **1006** and includes one or more input devices to receive input control commands to control operations of the manipulators **140**, instruments **150**, rails assembly **120**, and/or table assembly **101**. Such input devices can include but are not limited to, for example, telepresence input devices, triggers, grip input devices, buttons, switches, pedals, joysticks, trackballs, data gloves, trigger-guns, gaze detection devices, voice recognition devices, body motion or presence sensors, touchscreen technology, or any other type of device for registering user input. In some cases, an input device can be provided with the same degrees of freedom as the associated instrument that they control, and as the input device is actuated, the instrument, through drive inputs from the manipulator assembly, is controlled to follow or mimic the movement of the input device, which can provide the user a sense of directly controlling the instrument.

Telepresence input devices can provide the operator with telepresence, meaning the perception that the input devices are integral with the instrument. The user input and feedback system **1004** can also include feedback devices, such as a display device (not shown) to display images (e.g., images of the workspace as captured by one of the instruments **1010**), haptic feedback devices, audio feedback devices, other graphical user interface forms of feedback, etc.

[0085] The control system **1006** can control operations of the system **100**. In particular, the control system **1006** can send control signals (e.g., electrical signals) to the table assembly **101**, rail assembly **120**, manipulators **140**, and/or instruments **150** to control movements and/or other operations of the various parts. In some embodiments, the control system **1006** can also control some or all operations of the user input and feedback system **1004**, the auxiliary system **1008**, or other parts of the system **100**. The control system **1006** can include an electronic controller to control and/or assist a user in controlling operations of the manipulator assembly **1001**. The electronic controller includes processing circuitry configured with logic for performing the various operations. The logic of the processing circuitry can include dedicated hardware to perform various operations, software (machine readable and/or processor executable instructions) to perform various operations, or any combination thereof.

[0086] In configurations in which the logic includes software, the processing circuitry can include a processor to execute the software instructions and a memory device that stores the software. The processor can include one or more processing devices capable of executing machine readable instructions, such as, for example, a processor, a processor core, a central processing unit (CPU), a controller, a microcontroller, a system-on-chip (SoC), a digital signal processor (DSP), a graphics processing unit (GPU), etc. In cases in which the processing circuitry includes dedicated hardware, in addition to or in lieu of the processor, the dedicated hardware can include any electronic device that is configured to perform specific operations, such as an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Complex Programmable Logic Device (CPLD), discrete logic circuits, a hardware accelerator, a hardware encoder, etc. The processing circuitry can also include any combination of dedicated hardware and processor plus software.

[0087] Differing degrees of user control versus autonomous control can be utilized in the system **100**, and embodiments disclosed herein can encompass fully user-controlled systems, fully autonomously-controlled systems, and systems having any combination of user and autonomous control. For operations that are user-controlled, the control system **1006** generates control signals in response to receiving a corresponding user input command via the user input and feedback system **1004**. For operations that are autonomously controlled, the control system **1006** can execute pre-programmed logic (e.g., a software program) and can determine and send control commands based on the programming (e.g., in response to a detected state or stimulus specified in the programming). In some systems, some operations can be user controlled and others autonomously controlled. Moreover, some operations can be partially user controlled and partially autonomously

controlled—for example, a user input command can initiate performance of a sequence of events, and then the control system **1006** can perform various operations associated with that sequence without needing further user input.

[0088] In some embodiments, the control system **1006** can control positions of the manipulators **140** and the positions or configurations of the rail assembly **120** and/or the table assembly **101** by sending electrical signals to drive actuators that move the manipulators **140**, the rail assembly **120**, and/or the table assembly **101**. In some embodiments, the control system **1006** can determine states or configurations of the table assembly **101** and change a configuration of the rail assembly **120** based on the determined state or configuration of the table assembly **101**. For example, in response to determining that the table assembly **101** is going to be transitioned to a state/configuration in which interference between the rail assembly **120** and parts of the table assembly **101** might occur, the control system **1006** can automatically move the rail assembly **120** into a predetermined configuration in which the rail assembly **120** will not interfere with the table assembly **101**. For example, if the second end portion **103_3** is going to be moved into a declined position, the control system **1006** can move the rail **121** toward the first end portion **103_1** (e.g., the configuration shown in FIG. 1C) so that the rail **121** is out of the way of the second end portion **103_3**. The control system **1006** can determine that a transition to a given state/configuration of the table assembly **101** is occurring or is about to occur by, for example, detecting one or more sensor signals indicative that the transition is occurring (e.g., from a sensor configured to sense motion/position of a platform section **103** or other portion of the table assembly **101**) or by receiving input or signals associated with a request/command for the state transition to occur (e.g., in response to a user pressing a button to cause the state change).

[0089] The auxiliary system **1008** can include various auxiliary devices that can be used in operation of the system **100**. For example, the auxiliary system **1008** can include power supply units, auxiliary function units (e.g., functions such as irrigation, evacuation, energy supply, illumination, sensors, imaging, etc.). As one example, in a system **100** for use in a medical procedure context, the auxiliary system **1008** can include a display device for use by medical staff assisting a procedure, while the user operating the input devices can utilize a separate display device that is part of the user input and feedback system **1004**. As another example, in a system **100** for use in a medical context, the auxiliary system **1008** can include flux supply units that provide surgical flux (e.g., electrical power) to instruments **150**. An auxiliary system **1008** as used herein can thus encompass a variety of components and does not need to be provided as an integral unit.

[0090] Turning now to FIGS. 3-5D, a table-mounted manipulator system **200** (“system **200**”) in accordance with another embodiment is illustrated. The system **200** can be used as the system **100**, and some components of the system **200** can be used as components of the system **100** described above. Thus, the descriptions of the components of the system **100** above are applicable to the related components of the system **200**, and duplicative descriptions of these components are omitted below. The related components of the systems **100** and **200** are given reference numbers having the same right-most two digits—for example, **105** and **205**. Although the system **200** is one embodiment of the system **100**, the system **100** is not limited to the system **200**.

[0091] As shown in FIGS. 3-5D, the system **200** includes a table assembly **201**, two rail assemblies **220** coupled to the table assembly, and multiple manipulators **240** coupled to the rail assemblies **220**. Each manipulator **240** carries one or more instruments (not illustrated), which can be removably or permanently mounted thereon, as described above. The system **200** also can include a control system (not illustrated), a user input and feedback system (not illustrated), and/or an auxiliary system (not illustrated) similar to those described above in relation to the system **100**. In some embodiments, the system **200** is configured as a computer-assisted, teleoperable medical system. In other embodiments, the system **200** is configured as a teleoperable system for use in non-medical contexts.

[0092] As shown in FIGS. 3 and 4A, the table assembly **201** includes a platform assembly **210**

configured to support the patient or inanimate workpiece, a support column **202** coupled to and supporting the platform assembly **210**, and a base **205** coupled to the support column **202**. The base **205** can contact the ground or other surface upon which the table assembly **201** rests, and in some embodiments the base **205** includes wheels **206** (or other mobility devices as described above) to allow movement of the system **200** along the ground or other surface. As shown in FIG. 4A, in some embodiments, the support column **202** includes a telescoping support column that can raise or lower the platform assembly **210**.

[0093] The platform assembly **210** includes multiple platform sections **203** configured to support the patient or workpiece. In particular, in the embodiment illustrated in FIGS. 3-5, the platform assembly **210** includes first end section **203_1** ("head section **203_1**"), middle sections **203_2** and **203_3**, and second end section **203_4** ("foot section **203_4**"), which are arranged in series and movably coupled together via joints **207**. In some embodiments, the first end section **203_1** is configured to support a head of the patient, the second end section **203_4** is configured to support the feet and/or legs of the patient, and the more middle sections **203_2** and **203_3** are configured to support a torso and/or other portions of the patient. The joints **207** allow adjacent platform sections **203** to pivot relative to one another about rotation axes parallel to a lateral dimension **299** of the platform assembly **210** (e.g., parallel to a y-axis in the Figures).

[0094] FIG. 3 illustrates the platform assembly **210** in a neutral configuration in which all of the platform sections **203** are parallel to one another, and FIG. 4A illustrates the platform assembly **210** in an articulated configuration in which some of the platform sections **203** are oriented at non-zero angles relative to adjacent platform sections **203**. In some embodiments, some of the joints **207** also allow for other motion between adjacent platform sections **203**, such as relative translation along the longitudinal dimension **298** or relative rotation around a vertical axis parallel to a height dimension (i.e., the z-axis in the Figures), which is perpendicular to the lateral and longitudinal dimensions **299** and **298**. In some embodiments, the platform sections **203** include relatively rigid support portions **203b** and softer cushion portions **203a** attached to the support portions **203b**, with a surface of the cushion portions **203a** (i.e., the top surface in the orientation illustrated in FIG. 4) forming a support surface that contacts the patient or workpiece. In some embodiments, multiple platform sections **203** can share some components. For example, as illustrated in FIG. 3, the middle platform sections **203_2** and **203_3** can share the same cushion portion **203a** that extends across both platform sections **203_2** and **203_3**. The cushion portion **203a** shared by the platform sections **203_2** and **203_3** can bend when the platform sections **203_2** and **203_3** are articulated relative to one another, as shown in FIG. 4A.

[0095] In addition to moving individual platform sections **203** relative to adjoining platform sections **203**, the platform assembly **210** as a whole is movable relative to the support column **202**. In some embodiments, the middle section **203_3** is coupled to the support column **202** by one or more joints (not illustrated), providing for motion between the middle section **203_3** and the column **202**. The other platform sections **203_1**, **203_2**, and **203_4** are coupled (directly or indirectly) to the middle section **203_3**, and thus as the middle section **203_3** moves relative to the support column **202** the platform assembly **210** as a whole moves relative to the support column **202**.

[0096] In some embodiments, the motion of the middle section **203_3** (and hence platform assembly **210** as a whole) relative to the support column **202** includes pivoting (tilting) about a horizontal axis parallel to the lateral dimension **299** (e.g., a pitch degree of freedom of motion), as shown in FIG. 4A. In some embodiments, other degrees of freedom of motion are provided between the middle section **203_3** and the support column **202**, including pivoting (tilting) about a horizontal axis parallel to the longitudinal dimension **298** (e.g., a roll degree of freedom of motion), rotating about a vertical axis (e.g., a yaw degree of freedom of motion), and/or translation along the longitudinal and/or lateral dimensions **299** or **298**. As shown in FIG. 3, in the neutral configuration of the platform assembly **210**, the platform assembly **210** is parallel to the ground (or other surface

upon which the system **200** rests)—in particular, the support surfaces of the platform sections **203** are parallel to the ground (or other surface).

[0097] As shown in FIGS. **3-5D**, the platform assembly **210** also includes a number (one or more) of accessory rails **204** attached to side surfaces of the support portions **203b** of the platform sections **203**.

[0098] As noted above, the system **200** includes multiple manipulators **240**. In the embodiment illustrated in FIG. **3**, four manipulators **240** are present, with two manipulators **240** on each longitudinal side of the platform assembly **210** (i.e., two manipulators **240** are mounted to a rail assembly **220** along a first longitudinal side of the platform assembly **210** and two manipulators **240** are mounted to a rail assembly **220** along a second longitudinal side of the platform assembly **210**, wherein the second longitudinal side is opposite to the first longitudinal side). In other embodiments, more or fewer manipulators **240** can be mounted to rails **221**, such as one, two, three, or more manipulators per longitudinal side.

[0099] In FIG. **3**, the manipulators **240** are shown in a stowed state under end section **203_4**, while in FIGS. **4A-5D** the manipulators **240** are shown in various deployed states (only the two manipulators **240** disposed on one side of the platform assembly **210** are visible in FIGS. **4A-5D**). The deployed states include states in which the manipulators **240** are not stowed, e.g., distal ends of the manipulators **240** are positioned at or above the level of the platform assembly **110**, and can include a variety of configurations and positions of the manipulators **240** including but not limited to those shown in FIGS. **4A-5C**. The manipulators **240** include a number of links and joints, as well as an instrument manipulator mount **241** configured to receive an instrument removably mounted thereon and including an interface with output couplers to transfer driving force or other inputs to the instrument.

[0100] As shown in FIGS. **3-5D**, the manipulators **240** are coupled to the table assembly **201** via the rail assemblies **220**. One rail assembly **220** is provided for each of the two longitudinally extending sides of the platform assembly **210**. The description below describes a single rail assembly **220** for ease of description, and the other rail assembly **220** can be configured similarly. As shown in FIGS. **4A** and **4B**, the rail assembly **220** includes a movable rail **221** (“rail **221**”), movable manipulator carriages **226** coupled to the rail **221** and to the manipulators **240** (at least one movable manipulator carriage **226** per manipulator **240**), and one or more rail mounting carriages **227** coupled to the rail **221** and the table assembly **201** (see FIG. **4B**).

[0101] Each movable manipulator carriage **226** couples a respectively corresponding one of the manipulators **240** to the rail **221** such that the manipulators **240** can translate relative to the rail **221** along a longitudinal dimension **297** of the rail **221**. In addition, each rail mounting carriage **227** couples the rail assembly **220** to the platform assembly **210** (e.g., to the middle section **203_3**) such that the rail **221** can translate relative to the platform assembly **210** and the support column **202** along the longitudinal dimension **297** of the rail **221**. In the embodiment of FIGS. **3-5D**, because the rail assembly **220** is coupled to the platform middle section **203_3**, the longitudinal dimension **297** of the rail **221** is parallel to the longitudinal dimension **298** of the platform assembly **210** regardless of how the platform assembly **210** is moved or oriented relative to the support column **202**.

[0102] As shown in FIG. **4B**, which depicts interior components of the rail **221**, the movable rail **221** includes first and second sets of engagement features **222** and **223** configured to engage with complementary engagement features of the carriages **226** and **227**, respectively, to movably couple the rail **221** to the carriages **226** and **227**. The first set of engagement features **222** is configured to engage with complementary engagement features of the movable manipulator carriages **226**, while the second set of engagement features **223** is configured to engage with complementary engagement features (not visible) of the rail mounting carriages **227**, similar to that described with reference to FIGS. **1A-2**.

[0103] The first and second sets of engagement features **222** and **223** each includes two tracks

including flanges and/or grooves extending along the longitudinal dimension **297**, and the complementary engagement features of the carriages **226** and **227** are configured to engage and ride along the flanges and/or in the grooves of the first and second sets of engagement features **222** and **223**, respectively. The tracks of the first set of engagement features are spaced apart from one another in the lateral dimension to provide increased stability to and strengthen the coupling with the movable manipulator carriages **226**. The second set of engagement features **223** is configured similarly. In some embodiments, the complementary engagement features of the carriages **226** and **227** include bearing devices (not illustrated) configured to reduce friction to facilitate easier translation, such as wheels, balls, plain bearing surfaces, etc.

[0104] As shown in FIG. **4B**, the rail assembly **220** further includes a first translation drive mechanism to provide relative translation between the rail **221** and the movable manipulator carriages **226** and thus the manipulators **240** supported by the rail **221**. The first translation drive mechanism includes a lead screw **230** extending along the longitudinal dimension **297** and fixed relative to the rail **221**, follower nuts **231** engaged with the lead screw **230**, and coupled to the carriages **226**, and actuators **232** coupled to the follower nuts **231**. The lead screw **230** includes an external threaded surface and the follower nuts **231** include a complementary internal threaded surface engaged with the external threaded surface of the lead screw **230** such that rotation of the follower nuts **231** relative to the lead screw **230** causes translation of the follower nuts **231** along the lead screw **230**.

[0105] The actuators **232** can be electric motors or other actuation devices (e.g., hydraulic, pneumatic, or other devices to provide a motive force) configured to drive rotation of a corresponding follower nut **231**. The actuators **232** are operably coupled to the corresponding follower nut **231**, for example, via a gear mechanism or other motion transfer mechanism that converts motion of the actuator **232** into rotation of the follower nut **231**. A follower nut **231** is coupled to a corresponding movable manipulator carriage **226** such that the follower nut **231** and carriage **226** are constrained to translate together (i.e., to remain translationally stationary relative to one another) but with the follower nut **231** being rotatable relative to the carriage **226** (i.e., rotatable around the lead screw **230**). A coupling mechanism (not visible in FIG. **4B**) couples the follower nut **231** to the carriage **226** and extends down from the follower nut **231** through an opening provided in a bottom wall of the rail **221** in the region between the two tracks of the first set of engagement features **222**. In some embodiments, instead of or in addition to the lead screw **230**, the first translation mechanism can include one or more ball screws, a rack-and pinion drive, a cable/pulley drive system (described in more detail below), or various other devices to actuate and impart translational movement between the carriage **226** and rail **221**.

[0106] In the embodiment of FIGS. **3-5**, the one or more rail mounting carriages **227** couple the rail **221** to the table assembly **201**, for example to the middle section **203_3** in the embodiment illustrated in FIGS. **3-5D**, such that the rail **221** can also translate relative to the platform assembly **210** and support column **202** in a direction of the longitudinal dimension **297** of the rail **221**. This translation between the rail **221** and the platform assembly **210** and/or support column **202** is provided at least in part by the rail **221** translating relative to the rail mounting carriages **227**. Specifically, the rail **221** can include a second translation drive mechanism including a second lead screw **235**, follower nut **236**, and actuator **237**, which components are configured and operate similarly to the lead screw **230**, follower nut **231**, and actuator **232** described above except for being coupled to the rail mounting carriages **227** instead of the movable manipulator carriages **226**. The actuator **237** drives rotation of the follower nut **236**, which causes translation of the follower nut **236** along the lead screw **235**, which in turn causes relative translation between the carriages **227** and the rail **221**. Because the carriages **227** are attached to the platform assembly **110** (or alternatively support column), the relative translation between the carriages **227** and the rail **221** causes translation of the rail **221** relative to the platform assembly **110** (i.e., in a reference frame fixed to the platform assembly **110** or to the ground, the carriages **227** are seen as stationary and the

rail **221** is seen as translating). In some embodiments, instead of or in addition to the lead screw **235**, the second drive translation mechanism can include one or more ball screws, a rack-and-pinion drive, a cable/pulley drive system (described in more detail below), or various other devices to actuate and impart translational movement between the carriage **227** and rail **221**.

[0107] FIGS. **8-10** depict another embodiment of translation drive mechanisms and components thereof that can be used with a rail assembly of various embodiments, such as for example, with rail assemblies **120** or **220**. FIG. **8A** shows a rail assembly **820** mounted to a table assembly **801** and FIG. **8B** shows a portion of a rail assembly **820** is shown in isolation, which can be used as rail assembly **120** or **220**. Aspects of the table assembly and manipulators are not shown and not described here, but can be similar to the configurations and components described above with reference to other embodiments, such as table assemblies **101**, **201**, **301** and respective components. The rail assembly **820** includes a movable rail **821**, two moveable manipulator carriages **826** carrying coupling portions **845** configured to mount and support manipulator arms (not shown) moveably mounted to the movable rail **821**, and a rail mounting carriage **827** configured to mount the moveable rail **821** to the table assembly **801** (e.g., either to the platform assembly or support column as described above with reference to other embodiments). The rail **821** includes engagement features on opposing sides that are complementary to and engageable with engagement features on the carriages **826** and **827**, similar to engagement features **122**, **123**, **222**, **223**, **125**, **128**, **228** described above, to allow the carriages to suspend from and ride along the rail **821**.

[0108] To drive translation of the moveable manipulator carriages **826** and rail **821** relative to each other, and the rail **821** and rail mounting carriage **827** relative to each other, translation drive mechanisms that include a cable, pulley, and capstan arrangement are used. FIGS. **9A** and **9B** are perspective views of FIG. **8B** with the outer housing structure of the rail **821** removed to better illustrate the translation drive mechanism components, with the views of FIGS. **9A** and **9B** being taken from opposite sides. Three translation drive mechanisms are shown, two of which are respectively operably coupled to drive relative translation of the carriages **826** (only one of the carriages **826** shown in FIGS. **8A** and **8B** is illustrated in FIGS. **9A** and **9B** for simplicity) and rail **821**, and one of which is operably coupled to drive relative translation of rail mounting carriage **827** and rail **821**. More specifically, due to the fixed mounting of the rail mounting carriage **827** relative to the platform assembly (not shown), the rail **821** moves relative to the rail mounting carriage **827**, and the moveable manipulator carriages **826** move relative to the rail **821**.

[0109] Each translation drive mechanism includes a cable **838_1**, **838_2**, **838_3** looped around a pair of pulleys **839_1**, **839_2**, **839_3** disposed at opposite end portions of the rail **821** along the longitudinal dimension **897**. The cables **838_1**, **838_2**, **838_3** further comprise opposite free end portions wrapped in opposite directions around a capstan **842_1**, **842_2**, **842_3** (third capstan **842_3** is not visible in FIG. **9A** because hidden by housing) oriented with its axis of rotation generally along the longitudinal dimension **897** of the rail **821**. An actuator **832_1**, **832_2**, **832_3** is operably coupled to each capstan **842_1**, **842_2**, **842_3** to drive rotation of the capstan **842_1**, **842_2**, **842_3**. The manipulator carriages **826** (only one of which is illustrated) and rail mounting carriage **827** are respectively coupled in a fixed manner to a continuous length portion of the cables **838_1**, **838_2**, **838_3** that extend between the pulleys **839_1**, **839_2**, **839_3** opposite to where the capstan **842_1**, **842_2**, **842_3** is located between the pulleys **839_1**, **839_2**, **839_3**.

[0110] Actuation of a capstan **842_1**, **842_2**, **842_3** by respective actuator **832_1**, **832_2**, **832_3** to rotate in a given direction causes one free end portion of the cable **838_1**, **838_2**, **838_3** to wrap further onto the capstan **842_1**, **842_2**, **842_3** and the other free end portion of the cable **838_1**, **838_2**, **838_3** to wrap off the capstan **842_1**, **842_2**, **842_3** (also referred to as paying in and paying out of the cable relative to the capstan). This in turn causes the continuous length portion of the cable between pulleys **839_1**, **839_2**, **839_3** to translate in a given direction, thereby causing translation in the same direction of the carriage **826**, **827** attached to the continuous length portion

of the cable **838_1**, **838_2**, **838_3**. Rotation of the capstan **842_1**, **842_2**, **842_3** in the direction opposite of the given direction reverses which free end portion of the cable **838_1**, **838_2**, **838_3** is wrapped onto and off of the capstan **842_1**, **842_2**, **842_3** and so too the direction of translation of the continuous length portion of the cable **838_1**, **838_2**, **838_3** and corresponding attached carriage **826**, **827**.

[0111] The actuators **832** can be electric motors or other actuation devices (e.g., hydraulic, pneumatic, or other devices to provide a motive force) configured to drive rotation of a corresponding shaft to which the respective capstan can be coupled.

[0112] As depicted in FIGS. **9A**, to conserve space for the three translation drive mechanisms housed in the rail **821**, each of the manipulator carriages **826** (only one of which is shown in FIG. **9A**, with two shown in FIGS. **8A** and **8B**) can be provided with a through-passage to allow the cable that a respective carriage **826** is not operably coupled to translate to pass freely through the carriage as it moves to drive the other carriage **826**. For example, carriage **826** shown in FIG. **9A** has a through-passage **829** through which cable **838_2** passes. Further, the actuator/capstan pairs for each drive mechanism can be respectively placed at opposite end portions of the rail **821** for the manipulator carriages **826** and at a middle portion of the rail mounting carriage **821** for the carriage **827**. The cables **838_1**, **838_2** and corresponding pulleys **839_1**, **839_2** operably coupled to drive the manipulator carriages **826** can also be placed in a lower portion of the rail **821**, while the cable **838_3** and pulleys **839_3** operably coupled to carriage **827** is placed above the cables **838_1**, **838_2**.

[0113] The actuators **832_1**, **832_2**, **832_3** and pulleys **839_1**, **839_2**, **839_3** can be mounted in a fixed position relative to the rail **821**, and in some embodiment so are the capstans **842_1**, **842_2**, **842_3**. In another embodiment, to further conserve space while maximizing the distance the continuous portion of a cable **838_1**, **838_2**, **838_3** can travel without the cable **838_1**, **838_2**, **838_3** unwrapping from the capstan **842_1**, **842_2**, **842_3**, a traveling capstan arrangement can be used.

[0114] FIG. **10** illustrates a detailed view showing an embodiment of a traveling capstan arrangement that can be used as a nonlimiting embodiment of the capstans of the translation drive mechanisms of FIGS. **8** and **9**. In the embodiment of FIG. **10**, the actuator **932** includes a drive shaft in the form of a ball spline shaft **933**. A capstan **942** is coupled to the ball spline shaft **933**. The capstan **942** includes a first portion **942'** and a second portion **942''** having outer threaded surfaces. The respective opposite free end portions of a cable **938** are secured to and wrap in opposite directions on the respective first portion **942'** and second portion **942''** of the capstan **942**. In this way, the portion of the cable **938** wrapped around first portion **942'** and the portion of cable **938** wrapped around second portion **942''** respectively wrap and unwrap from the capstan **942** oppositely in response to rotation of the capstan **942** in a given direction.

[0115] The balls spline shaft **933** coupling to the capstan **942** allows the capstan **942** to travel along the ball spline shaft **933** in differing directions depending on the direction of rotation of the ball spline shaft **933** as driven by actuator **932**. As the capstan **942** moves long the ball spline shaft **933**, the portions of the cable **938** can be paid in or out (wrap on or off) the respective portions **942'**, **942''** oppositely depending on the direction of travel. This can provide a relatively compact arrangement of the capstan **942** to be used while still providing a relatively large range of translational motion of the continuous portion of the cable **938** (not shown in FIG. **10**) and thus of the attached carriage.

[0116] To assist with guiding the cable **938** onto and off of the capstan **942** and routing the cable toward the pulleys associated with the cable **938** (such as pulleys **839** of FIGS. **8** and **9**, with pulleys not shown **10** FIG. **10**), a pair of redirecting pulleys **934** can be used. The redirecting pulleys **934** can be fixed in a housing (shown transparent) to which the actuator **932**, capstan **942**, and ball spline shaft **933** are fixedly coupled. Openings in the housing can permit the cable **938** to exit the housing and be routed in opposite directions to the pair of pulleys the cable is coupled to.

[0117] The traveling capstan arrangement of FIG. 10 can in various embodiments have a configuration and operation like that disclosed in WO 2022/192109, published Sep. 15, 2022, the entire contents of which are incorporated by reference herein. This is just one nonlimiting embodiment of a traveling capstan arrangement that can be used, and other traveling capstan arrangements are contemplated as within the scope of the present disclosure. In another embodiment, as noted above, it is contemplated that the capstan arrangement could be fixed, and if redirecting pulleys are employed, the redirecting pulleys can be moveable to align the cable at a desired fleet angle with the capstan.

[0118] FIGS. 5A-5D show the platform assembly 210 and the rail assembly 220 in various configurations illustrative of a range of motion provided by the rail assembly 220 and various ranges of motion of the platform assembly 210 in some embodiments. Those having ordinary skill in the art will appreciate that the various configurations illustrated in FIGS. 5A-5D are non-limiting and other configurations are attainable in view of the various degrees of freedom of the rail 221, platform assembly sections 203, carriages 226, 227, and manipulators 240. In FIG. 5A, the rail mounting carriage 227 is positioned at a middle portion of the rail 221, such that the rail 221 is situated near the middle sections 203_2 and 203_3 of the platform assembly 210, with some portion of the length of the rail 221 extending near section 203_4. In FIG. 5A, the platform assembly 210 is in a neutral position with the manipulators 240 in a deployed state.

[0119] In FIG. 5B, the rail mounting carriage 227 (not visible in FIG. 5B) is positioned at a head end portion of the rail 221 (toward the left end of the rail 221 in the figure), such that the rail 221 has translated to the right from FIG. 5A and is situated near the foot end portion of the table assembly 220 (under the middle section 203_3 and the foot section 203_4). FIG. 5B also illustrates optional declined and inclined configurations the middle section 203_2 can be placed in in this positioning of the rail 221 (the declined configuration is illustrated in dashed lines and the inclined configuration in solid). The positioning of the rail assembly 220 in the configuration of FIG. 5B allows the first end section 203_1 to be declined or inclined (not illustrated) relative to the middle section 203_2, and/or allows the middle section 203_2 to be declined or inclined relative to the middle section 203_3 as illustrated in FIG. 5B, without interference (collision) with the rail assembly 220. Moreover, in this configuration of the rail assembly 220, the manipulators 240 can be moved in a longitudinal direction beyond a foot end of the platform assembly 210 (i.e., beyond the foot section 203_4) and then in a lateral direction such that the manipulators 240 are positioned along the laterally extending side of the platform assembly 210, as shown in FIG. 5B, instead of along the longitudinally extending side, as shown in FIG. 5A. This allows the manipulators to be positioned out of the way of operations performed along the longitudinally extending side of the platform assembly 210, such as an operation of transferring a patient from a gurney to the platform assembly 210.

[0120] In FIG. 5C, the rail mounting carriage 227 is positioned at a right end portion of the rail 221, such that the rail 221 is near a left end portion of the platform assembly 210. This configuration of the rail assembly 220 can allow the second end section 203_4 to be declined relative to the middle section 203_3 without collision with the rail assembly 220. FIG. 5C also illustrates the second end section 203_4 in declined and neutral configurations.

[0121] In FIG. 5D, inclined and declined configurations of the platform assembly 210 as a whole are illustrated. These configurations can be achieved by rotating the middle section 203_3 relative to the support column about an axis parallel to the lateral dimension 299. As shown in FIG. 5D, the rail assembly 220 and the manipulators 240 move along with the platform assembly 210 in these embodiments because the rail assembly 220 is attached to the platform assembly 210. In other embodiments (not illustrated), the rail assembly 220 can be attached to the support column 202.

[0122] Thus, the rail assembly 220 provides a wide range of motion for the manipulators 240, which can allow the manipulators 240 to be positioned nearly anywhere along the longitudinal dimension 298 of the platform assembly 210, including positioning the manipulators 240 beyond or

outside of a head or foot end of the platform assembly such that the manipulators **240** can be clear from the entire longitudinal dimension **298** of the platform assembly. Moreover, not only does the rail assembly **220** provide a wide range of motion for the manipulators **240**, the rail assembly **220** does so while also minimizing interference between the rail assembly **220** and the platform assembly **210**, personnel, and other equipment. In particular, the rail assembly is movable between different positions as noted above based on the configuration of the platform assembly **210** and/or based on the needs of a particular operation such that the rail **221** is positioned out of the way of the portions of the platform assembly **210**, personnel, or other equipment as needed.

[0123] Furthermore, the rail assembly **220** is configured to remain within an envelope **208** defined by the platform assembly **210** throughout all motion of the rail **221** relative to the platform assembly **210** and/or support column **202**. This further reduces the degree of interference between the rail assembly **220** and other equipment, as will be described in greater detail below. The envelope **208** (see FIG. 3) is defined by the platform assembly **210** in a similar manner as the envelope **108**, and thus duplicative description thereof is omitted.

[0124] Turning now to FIGS. 6 and 7, an embodiment of a table-mounted manipulator system **300** (“system **300**”) is described below. The system **300** can be used as the system **100**, and some components of the system **300** can be used as components of the system **100** described above. Thus the descriptions of the components of the system **100** above are applicable to the related components of the system **300**, and duplicative descriptions of these components are omitted below. The related components of the systems **100** and **300** are given reference numbers having the same right-most two digits—for example, **105** and **305**. Although the system **300** is one embodiment of the system **100**, the system **100** is not limited to the system **300**.

[0125] As shown in FIGS. 3-5D, the system **300** includes a table assembly **301**, a rail assembly **320** coupled to the table assembly, and multiple manipulators **340** coupled to the rail assembly **320**. Each manipulator **340** carries one more instruments (not illustrated), which can be removably or permanently mounted thereon. The system **300** also can include a control system (not illustrated), a user input and feedback system (not illustrated), and/or an auxiliary system (not illustrated) similar to those described above in relation to the system **100**. In some embodiments, the system **300** is configured as a computer-assisted, teleoperable medical system. In other embodiments, the system **300** is configured as a teleoperable system for use in non-medical contexts.

[0126] As shown in FIGS. 6 and 7, the table assembly **301** includes a platform assembly **310** configured to support the patient or inanimate workpiece, a support column **302** coupled to and supporting the platform assembly **310**, and a base **305** coupled to the support column **302**. The base **305** can be configured to contact the ground or other surface upon which the table assembly **301** rests, and in some embodiments the base **305** includes wheels **306** to allow movement of the system **300** along the ground or other surface. In some embodiments the support column **302** includes a telescoping support that can raise or lower the platform assembly **310** relative to the ground or other surface.

[0127] The platform assembly **310** includes multiple platform sections **303** configured to support the patient or workpiece. In particular, in the embodiment illustrated in FIGS. 6 and 7 the platform assembly **310** includes first end section **303_1**, middle sections **303_2** and **303_3**, and second end section **303_4**, which are arranged in series and movably coupled together via joints **307**, similar to the platform assembly **210** described above. The joints **307** allow adjacent platform sections **303** to pivot relative to one another as described above in relation to platform assembly **210**. In some embodiments, some of the joints **307** can also allow for other motion between adjacent platform sections **303**, such as relative translation along the longitudinal dimension **398** or relative rotation around a vertical axis. In some embodiments, the platform sections **303** include relatively rigid support portions **303b** and softer cushion portions **303a** attached to the support portions **303b**, with a surface of the cushion portions **303a** (i.e., the top surface in the orientation illustrated in FIG. 7) forming a support surface that contacts the patient or workpiece. In some embodiments, multiple

platform sections **303** can share some components. For example, as illustrated in FIGS. **6** and **7**, the middle platform sections **303_2** and **303_3** can share the same cushion portion **303a** that extends across both platform sections **303_2** and **303_3**.

[0128] In addition to moving individual platform section **303** relative to adjoining platform sections **303**, the platform assembly **310** as a whole is movable relative to the support column **302**. In some embodiments, the middle section **303_3** is coupled to the support column **302** by one or more joints (not illustrated), providing for motion between the middle section **303_3** and the column **302**. In some embodiments, the motions of the platform assembly **310** can be similar to those of the platform assembly **210** described above, and thus duplicative description thereof is omitted.

[0129] As shown in FIGS. **6** and **7**, the platform assembly **310** also optionally includes a number of accessory rails **304** attached to side surfaces of the support portions **303b** of the platform sections **303**.

[0130] As noted above, the system **300** includes multiple manipulators **340**. In the embodiment illustrated in FIGS. **6** and **7**, three manipulators **340** are present. In other embodiments, more or fewer manipulators **340** can be used. In the embodiment of FIGS. **6** and **7** all three manipulators **340** are positioned on the same longitudinally extending side of the platform assembly **310** and are coupled to the same rail assembly **320**. In other embodiments (not illustrated) of the system **300**, multiple rail assemblies **320** can be included (e.g., one on each longitudinally extending side of the platform assembly **310**, with a first rail assembly **320** on a first longitudinally extending side of the platform assembly **310** and a second rail assembly **320** on a second longitudinally extending side of the platform assembly **310**, wherein the second longitudinally extending side is opposite to the first longitudinally extending side). When multiple rail assemblies **320** are included, each rail assembly **320** can have one or more manipulators **340** coupled thereto (e.g., one, two, three, or more manipulators **340** can be coupled to each rail assembly **320**). The manipulators **340** include a number of links and joints, as well as an instrument manipulator mount **341** configured to receive an instrument removably mounted thereon and including an interface with output couplers to transfer driving force or other inputs to the instrument.

[0131] As shown in FIGS. **6** and **7**, the manipulators **340** are coupled to the table assembly **301** via the rail assembly **320**. As shown in FIGS. **6** and **7**, the rail assembly **320** includes a movable rail **321** (“rail **321**”), movable manipulator carriages **326** coupled to the rail **321** and to the manipulators **340** (at least one movable manipulator carriage **326** per manipulator **340**), and one or more rail mounting carriages **327** coupled to the rail **321** and the table assembly **301**.

[0132] Each movable manipulator carriage **326** couples a respectively corresponding one of the manipulators **340** to the rail **321** such that the manipulators **340** can translate relative to the rail **321** along a longitudinal dimension **397** of the rail **321**. In addition, each rail mounting carriage **327** couples the rail assembly **320** to the platform assembly **310** (e.g., to the middle section **303_3**) such that the rail **321** can translate relative to the platform assembly **310** and the support column **302** along the longitudinal dimension **397** of the rail **321**. In the embodiment of FIGS. **6** and **7**, because the rail assembly **320** is coupled to the platform section **303_3**, the longitudinal dimension **397** of the rail **321** is parallel to the longitudinal dimension **398** of the platform assembly **310** regardless of how the platform assembly **310** is moved relative to the support column **302**.

[0133] As shown in FIGS. **6** and **7**, the movable rail **321** includes first and second sets of engagement features **322** and **323** configured to engage with complementary engagement features of the carriages **326** and **327**, respectively, to movably couple the rail **321** to the carriages **326** and **327**. The first and second sets of engagement features **322** and **323** each includes tracks including flanges and/or grooves extending along the longitudinal dimension **397**, and the complementary engagement features of the carriages **326** and **327** are configured to engage and ride along the flanges and/or in the grooves of the first and second sets of engagement features **322** and **323**, respectively. In some embodiments, the complementary engagement features of the carriages **326** and **327** can include bearing devices (not illustrated) configured to reduce friction to facilitate

easier translation, such as wheels, balls, plain bearing surfaces, etc.

[0134] In some embodiments, the carriages **326** and **327** are manually movable relative to the rail **321**. In other embodiments, the carriages **326** and **327** include actuators (not illustrated) to drive motion of the carriages **326** and **327** relative to the rail **321**. In some embodiments, the actuators can be housed within the carriages **326** and **327**, unlike the rail assemblies **220**, **820** in which the actuators were housed within the rail **221**. The actuators can, for example, include motors configured to drive rotation of gears and/or wheels engaged with the engagement features **322** and **323** to cause translation of the carriages **326** and **327** relative to the rail **321**.

[0135] In some embodiments, the rail mounting carriages **327** are coupled to the middle section **303_3** via a fixed rail **324** attached to the middle section **303_3**, as shown in FIGS. **6** and **7**. For example, in some embodiments a first end of the carriage **327** is movably coupled to the fixed rail **324**, while a second end of the carriage **327** is movably coupled to the movable rail **321** as described above. This allows the rail mounting carriage **327** to not only move relative to the rail **321**, but also to move relative to the platform assembly **310**. Thus, in these embodiments, translation of the rail **321** relative to the table assembly **301** can occur through either one of (or both of) relative translation between the carriages **327** and the table assembly **301** (i.e., relative to the fixed rail **324**) and relative translation between the carriages **327** and rail **321**. The movability of the rail mounting carriages **327** relative to the table assembly **301** can allow for a greater range of motion of the manipulators **340** for a given size of rail **321** and/or a smaller rail **321** to be used for a given range of motion.

[0136] The rail assembly **320** and the platform assembly **310** can be movable through ranges of motion similar to those already described above in relation to the system **200**, and thus duplicative description thereof is omitted. Thus, the rail assembly **320** provides a wide range of motion for the manipulators **340**, which can allow the manipulators **340**, where they attach to the carriages **326**, to be positioned nearly anywhere along the length of the platform assembly **310**. Moreover, not only does the rail assembly **320** provide a wide range of motion for the manipulators **340**, it does so while also minimizing interference between the rail assembly **320** and the platform assembly **310**, personnel, and other equipment. In particular, the rail assembly is movable between different positions as noted above based on the configuration of the platform assembly **310** and/or based on the needs of a particular operation such that the rail **321** is positioned out of the way of the portions of the platform assembly **310**, personnel, or other equipment as needed.

[0137] Furthermore, the rail assembly **320** is configured to remain within an envelope **308** defined by the platform assembly **310** throughout all motion of the rail **321** relative to the platform assembly **310** and/or support column **302**. This further reduces the degree of interference between the rail assembly **320** and other equipment. The envelope **308** (see FIG. **7**) is defined by the platform assembly **310** in a similar manner as the envelope **108**, and thus duplicative description thereof is omitted.

[0138] The embodiments described herein may be well suited for use in any of a variety of medical procedures, as described above. Such procedures could be performed, for example, on human patients, animal patients, human cadavers, animal cadavers, and portions or human or animal anatomy. Medical procedures as contemplated herein include any of those described herein and include, for non-surgical diagnosis, cosmetic procedures, imaging of human or animal anatomy, gathering data from human or animal anatomy, training medical or non-medical personnel, and procedures on tissue removed from human or animal anatomies (without return to the human or animal anatomy). Even if suitable for use in such medical procedures, the embodiments can also be used for benchtop procedures on non-living material and forms that are not part of a human or animal anatomy. Moreover, some embodiments are also suitable for use in non-medical applications, such as industrial robotic uses, and sensing, inspecting, and/or manipulating non-tissue work pieces. In non-limiting embodiments, the techniques, methods, and devices described herein can be used in, or be part of, a computer-assisted surgical system employing robotic

technology such as the da Vinci® Surgical Systems commercialized by Intuitive Surgical, Inc., of Sunnyvale, California. Those skilled in the art will understand, however, that aspects disclosed herein can be embodied and implemented in various ways and systems, including manually operated instruments and computer-assisted, teleoperated systems, in both medical and non-medical applications. Reference to the daVinci® Surgical Systems are illustrative and not to be considered as limiting the scope of the disclosure herein.

[0139] As used herein and in the claims, terms such as computer-assisted manipulator system, teleoperable manipulator system, or the like should be understood to refer broadly to any system including one or more controllable kinematic structures (“manipulators”) that are movable and controllable at least in part through the aid of an electronic controller (with or without human inputs). Such systems are occasionally referred to in the art and in common usage as robotically assisted systems or robotic systems. Such systems include systems that are controlled by a user (for example through teleoperation), by a computer automatically (so-called autonomous control), or by some combination of these. In examples in which a user controls at least some of the operations of the manipulator, an electronic controller (e.g., a computer) can facilitate or assist in the operation. The term “computer” as used in “computer-assisted manipulator systems” refers broadly to any electronic control device for controlling, or assisting a user in controlling, operations of the manipulator, and is not intended to be limited to things formally defined as or colloquially referred to as “computers.” For example, the electronic control device in a computer-assisted manipulator system could range from a traditional “computer” (e.g., a general-purpose processor plus memory storing instructions for the processor to execute) to a low-level dedicated hardware device (analog or digital) such as a discrete logic circuit or application specific integrated circuit (ASIC), or anything in between. Further, manipulator systems can be implemented in a variety of contexts to perform a variety of procedures, both medical and non-medical. Thus, although some examples described in greater detail herein are focused on a medical context, the devices and principles described herein are also applicable to other contexts, such as industrial manipulator systems.

[0140] It is to be understood that both the general description and the detailed description provide example embodiments that are explanatory in nature and are intended to provide an understanding of the present disclosure without limiting the scope of the present disclosure. Various mechanical, compositional, structural, electrical, and operational changes can be made without departing from the spirit and scope of this description and the claims. In some instances, well-known circuits, structures, and techniques have not been shown or described in detail in order not to obscure the embodiments. Like numbers in two or more figures represent the same or similar elements.

[0141] Further, the terminology used herein to describe aspects of the invention, such as spatial and relational terms, is chosen to aid the reader in understanding example embodiments of the invention but is not intended to limit the invention. For example, spatial terms—such as “beneath”, “below”, “lower”, “above”, “upper”, “proximal”, “distal”, “up”, “down”, and the like—are used herein to describe directions or one element's or feature's spatial relationship to another element or feature as illustrated in the figures. These spatial terms are used relative to the figures and are not limited to a particular reference frame in the real world. Thus, for example, the direction “up” in the figures does not necessarily have to correspond to an “up” in a world reference frame (e.g., away from the Earth's surface). Furthermore, if a different reference frame is considered than the one illustrated in the figures, then the spatial terms used herein may need to be interpreted differently in that different reference frame. For example, the direction referred to as “up” in relation to one of the figures may correspond to a direction that is called “down” in relation to a different reference frame that is rotated 180 degrees from the figure's reference frame. As another example, if a device is turned over 180 degrees in a world reference frame as compared to how it was illustrated in the figures, then an item described herein as being “above” or “over” a second item in relation to the Figures would be “below” or “beneath” the second item in relation to the world reference frame. Thus, the same spatial relationship or direction can be described using

different spatial terms depending on which reference frame is being considered. Moreover, the poses of items illustrated in the figure are chosen for convenience of illustration and description, but in an implementation in practice the items can be posed differently.

[0142] In addition, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context indicates otherwise. And, the terms “comprises”, “comprising”, “includes”, and the like specify the presence of stated features, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups. Components described as coupled can be electrically or mechanically directly coupled, or they can be indirectly coupled via one or more intermediate components, unless specifically noted otherwise. Mathematical and geometric terms are not necessarily intended to be used in accordance with their strict definitions unless the context of the description indicates otherwise, because a person having ordinary skill in the art would understand that, for example, a substantially similar element that functions in a substantially similar way could easily fall within the scope of a descriptive term even though the term also has a strict definition.

[0143] Elements and their associated aspects that are described in detail with reference to one embodiment can be, whenever practical, included in other embodiments in which they are not specifically shown or described. For example, if an element is described in detail with reference to one embodiment and is not described with reference to a second embodiment, the element may nevertheless be claimed as included in the second embodiment.

[0144] As used herein, “proximal” and “distal” are spatial/directional terms that describe locations or directions based on their relationship to the two ends of a kinematic chain. “Proximal” is associated with the end of the kinematic chain that is closer to the base or support of the chain, while “distal” is associated with the opposite end of the kinematic chain, which often includes an end effector of an instrument. When used in to refer to locations or to portions of a component, proximal and distal indicate the relative positions of the locations or portions relative to the base of the chain, with the proximal location or portion being closer to the base (closer here referring to proximity along the kinematic chain, rather than absolute distance). When used to refer to a direction, “proximal” refers to directions that point generally from a given location along a kinematic chain towards a more proximal location along the kinematic chain, and “distal” refers to directions that point from the given location towards a more distal location along the kinematic chain.

[0145] Unless otherwise noted herein or implied by the context, when terms of approximation such as “substantially,” “approximately,” “about,” “around,” “roughly,” and the like, are used in conjunction with a stated numerical value, property, or relationship, such as an end-point of a range or geometric properties/relationships (e.g., parallel, perpendicular, straight, etc.), this should be understood as meaning that mathematical exactitude is not required for the value, property, or relationship, and that instead a range of variation is being referred to that includes but is not strictly limited to the stated value, property, or relationship. In particular, the range of variation around the stated value, property, or relationship includes at least: any inconsequential variations; those variations that are typical in the relevant art for the type of item in question due to manufacturing or other tolerances; and/or variations that are within $\pm 5\%$ of the stated value, property, or relationship unless indicated otherwise.

[0146] As used herein, “transverse” refers to a positional relationship of two items in which one item is oriented crosswise at an angle relative to the other item, such as being substantially or generally perpendicular to the other item. As used herein, “transverse” includes, but does not require, an exactly perpendicular relationship.

[0147] Further modifications and alternative embodiments will be apparent to those of ordinary skill in the art in view of the disclosure herein. For example, the devices and methods can include additional components or steps that were omitted from the diagrams and description for clarity of

operation. Accordingly, this description is to be construed as illustrative only and is for the purpose of teaching those skilled in the art the general manner of carrying out the present teachings. It is to be understood that the various embodiments shown and described herein are to be taken as exemplary. Elements and materials, and arrangements of those elements and materials, can be substituted for those illustrated and described herein, parts and processes can be reversed, and certain features of the present teachings can be utilized independently, all as would be apparent to one skilled in the art after having the benefit of the description herein. Changes can be made in the elements described herein without departing from the spirit and scope of the present teachings and following claims.

[0148] It is to be understood that the particular examples and embodiments set forth herein are non-limiting, and modifications to structure, dimensions, materials, and methodologies can be made without departing from the scope of the present teachings.

[0149] Other embodiments in accordance with the present disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with the following claims being entitled to their fullest breadth, including equivalents, under the applicable law.

Claims

1. A teleoperable manipulator system, comprising: a table assembly comprising: a support column; and a platform assembly coupled to the support column and configured to support a body, wherein the platform assembly comprises lateral and longitudinal dimensions defining outer lateral and longitudinal boundaries of the platform assembly; a rail coupled to the table assembly, the rail being translatable relative to the support column along a longitudinal dimension of the rail; and one or more manipulators coupled to the rail and translatable relative to the rail along the longitudinal dimension of the rail, wherein the rail is configured to remain within at least the outer lateral boundaries of the platform assembly throughout all motion of the rail relative to the support column.
2. The system of claim 1, wherein the rail is configured to remain within the outer longitudinal boundaries of the platform assembly throughout all motion of the rail relative to the support column.
3. The system of claim 1, wherein the platform assembly comprises a plurality of sections movable relative to one another to change a configuration of the platform assembly between a plurality of configurations, and wherein the rail is movable between a plurality of positions to allow the platform assembly to move between the plurality of configurations without interference from the rail.
4. The system of claim 3, wherein the plurality of sections comprise a first end section, one or more middle sections, and a second end section consecutively positioned along the longitudinal dimension of the platform assembly, and wherein each of the first and second end sections are independently pivotable relative to the one or more middle sections.
5. The system of claim 3, further comprising: one or more first carriages, each first carriage movably coupling one of the manipulators to the rail, each first carriage being drivable to cause translation of the manipulator coupled thereto relative to the rail along the longitudinal dimension of the rail; and one or more second carriages movably coupling the rail to the table assembly, each second carriage being drivable to cause translation of the rail relative to the platform assembly and the support column.
6. The system of claim 5, a control system communicably connected to the first and second carriages, the control system being configured to drive the first and second carriages to position the rail and the manipulators.

7. The system of claim 3, wherein the plurality of sections comprise a first section and a second section positioned adjacent to and pivotable relative to the first section, wherein in a first state the second section is declined relative to the first section and at least a portion of the second section occupies a given region of space, wherein the rail is translatable relative to the support column between a first position and a second position, in the first position a portion of the rail occupies the given region of space, in the second position the rail does not occupy the given region of space.
8. The system of claim 7, further comprising: a control system configured to, in response to determining a transition to the first state is going to occur, cause the rail to move from the first position to the second position.
9. The system of claim 1, further comprising: one or more carriages, each carriage movably coupling one of the manipulators to the rail, each carriage being drivable in translation along the longitudinal dimension of the rail so as to cause translation of the manipulator coupled thereto relative to the rail along the longitudinal dimension of the rail.
10. (canceled)
11. The system of claim 9, further comprising a drive mechanism operably coupled to drive translation of the one or more carriages along the longitudinal dimension of the rail, wherein the drive mechanism comprises: a lead screw; one or more follower nuts engaged with the lead screw and coupled with respectively corresponding ones of the one or more carriages; and one or more actuators operably coupled to the follower nuts and configured to rotate the one or more follower nuts relative to the lead screw.
12. The system of claim 9, further comprising a drive mechanism operably coupled to drive translation of the one or more carriages along the longitudinal dimension of the rail, wherein the drive mechanism comprises: one or more cables coupled with respectively corresponding ones of the one or more carriages; one or more capstans, wherein the one or more cables are at least partially wrapped around the one or more capstans, respectively; and one or more actuators respectively operably coupled to the one or more capstans and configured to rotate the one or more capstans, wherein the one or more cables are configured to be moved in translation in response to rotation of the one or more capstans.
13. The system of claim 12, wherein the one or more carriages comprise a first carriage and a second carriage and the system further comprises a rail mounting carriage movably coupling the rail to the platform assembly, wherein the one or more cables comprise a first cable coupled with the first carriage, a second cable coupled with the second carriage, and a third cable coupled with the rail mounting carriage, wherein the one or more capstans comprises a first capstan, a second capstan, and a third capstan, wherein the first cable is at least partially wrapped around the first capstan, the second cable is at least partially wrapped around the second capstan.
14. The system of claim 11, further comprising: one or more rail mounting carriages movably coupling the rail to the platform assembly, the one or more rail mounting carriages being drivable to cause translation of the rail relative to the platform assembly and the support column.
15. The system of claim 14, further comprising: a second lead screw; one or more second follower nuts engaged with the second lead screw and coupled with respectively corresponding ones of the one or more rail mounting carriages; and one or more second actuators operably coupled to the second follower nuts to drive translation of respectively corresponding ones of the one or more rail mounting carriages relative to the rail by driving rotation of the respectively corresponding second follower nuts relative to the second lead screw.
16. The system of claim 14, further comprising: a fixed rail coupled to the platform assembly or to the support column, wherein the one or more rail mounting carriages are movably coupled to the fixed rail and are translatable relative to the platform assembly along the fixed rail.
17. The system of claim 1, wherein the platform assembly is tiltable relative to the support column; and the rail is configured to tilt along with the platform assembly relative to the support column.
18. The system of claim 17, wherein the platform assembly comprises a plurality of sections

movable relative to one another to change a configuration of the platform assembly between a plurality of configurations, wherein a first section of the plurality of sections is coupled to the support column and is tiltable relative to the support column, and wherein the rail is coupled to the first section.

19. The system of claim 1, further comprising: an accessory rail coupled to a longitudinal side of the platform assembly, the accessory rail configured to mount auxiliary medical equipment, wherein the outer lateral boundaries of the platform assembly are defined by outermost edges of the platform assembly along the longitudinal side excluding the accessory rail.

20. The system of claim 1, further comprising: a rail mounting carriage coupled to the platform assembly or to the support column, wherein the rail is movably coupled to the rail mounting carriage and is configured to move along the rail mounting carriage to cause translation of the rail relative to the support column along the longitudinal dimension of the rail; and a drive mechanism operably coupled to drive movement of the rail along the rail mounting carriage, wherein the drive mechanism comprises a lead screw drive mechanism or a capstan drive mechanism.

21.-23. (canceled)

24. A rail assembly for mounting manipulators to a table assembly, the rail assembly comprising: a rail having a longitudinal dimension; a rail mounting carriage movably engaged with the rail and configured to couple the rail to a table assembly; first and second manipulator mounting carriages movably engaged with the rail at an opposite side of the rail as the rail mounting carriage; and a drive mechanism housed in an interior of the rail, the drive mechanism comprising: a first capstan assembly, a first cable operably coupled to the first capstan assembly and to the first manipulator carriage, the first cable passing through the second manipulator carriage a second capstan assembly, a second cable operably coupled to the second capstan assembly and the second manipulator carriage, the second cable passing through the first manipulator carriage a third capstan assembly, a third cable operably coupled to the third capstan assembly and the rail mounting carriage, wherein the first, second, and third capstan assemblies are arranged in a series along the longitudinal dimension of the rail.
